Tampere University

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Mitochondria — Powerhouse of the Cell... and the Brain Tumor?

Computational Analysis of Apoptosis and Mitochondrial Respiratory Complexes in Classical Glioblastoma Transcriptome

Bachelor's Thesis
Engineering and Natural Sciences
September 2026

ABSTRACT

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The abstract is a concise, self-containing one-page description of the work: what was the problem, what was done, and what are the results. Do not include charts or tables in the abstract. Put the abstract in the primary language of your thesis first and then the translation when that is needed. International students do not need to include an abstract in Finnish.

Keywords: glioblastoma, apoptosis, mitochondria, respiratory complex, transcriptomics

The originality of this thesis has been checked using the Turnitin Originality service.

PREFACE

> *"DNA neither cares or knows. DNA just is. And we dance to its music."*
>
> — Richard Dawkins

This thesis was written as part of Tampere University's Bachelor's Programme in Science and Engineering and was begun in early May 2026. The work was supervised by Dr. Meenakshisundaram Kandhavelu, who inspired my interest in bioinformatics and provided me the opportunity to develop it further in his Molecular Signaling Lab.

This work also marks an important milestone not only for myself but also for my family.

I would like to thank my friends from various parts of the world for their encouragement, insight, and support throughout the writing process — both online and in-person. Finally, I express my deepest gratitude to my parents, V. and K., whose support made this achievement possible. The completion of this thesis — and the degree as a whole — would not have been possible without any of them.

In Tampere on 5. September 2026,
Nam Từ

Contents

GLOSSARY

DEG	(Significantly) differentially expressed genes
EGFR	Italic words denote genes
GBM	Glioblastoma multiforme

USE OF ARTIFICIAL INTELLIGENCE IN THIS WORK

Artificial intelligence (AI) has been used in generating this work:

- ☒ Yes
- ☐ No

I hereby declare, that the AI-based applications used in generating this work are as follows:

Application	**Version**
ChatGPT	GPT-5.5

PURPOSE OF THE USE OF AI

Explain here *in detail*, for which purpose and how AI was utilized in writing this thesis.

ChatGPT was used mainly for debugging code during analysis of multiomics.

PARTS OF THIS WORK, WHERE AI WAS USED

List here all chapters, sections, subsections, tables, figures and so forth, that were generated by an AI, or that an AI had a hand in generating.

ACKNOWLEDGEMENT OF RISKS

I hereby acknowledge, that as the author of this work, I am fully responsible for the contents presented in this thesis. This includes the parts that were generated by an AI, in part or in their entirety. I therefore also acknowledge my responsibility in the case, where use of AI has resulted in ethical guidelines being breached.

1 INTRODUCTION

Glioblastoma multiforme (GBM) is the most aggressive primary brain tumour in adults and remains associated with poor patient survival despite advances in treatment [1]. A major challenge in understanding and treating GBM is its substantial molecular heterogeneity. Molecular analyses have identified distinct GBM subtypes characterised by different transcriptional profiles and biological properties, which can influence tumour progression, treatment response, and patient prognosis [2]. Understanding these molecular differences is therefore important for characterising the biological mechanisms underlying GBM.

Advances in high-throughput transcriptomic technologies have enabled comprehensive characterisation of gene expression patterns in cancer. Transcriptomic data can be used to identify genes whose expression differs between tumour and healthy tissue and to investigate the biological processes associated with these changes [3]. However, lists of differentially expressed genes alone provide limited information about the biological mechanisms underlying the observed changes. Functional enrichment analysis can therefore be used to identify biological processes that are disproportionately represented among differentially expressed genes, providing a biological context for transcriptomic alterations [4].

Cell death is an important biological process in cancer, and dysregulation of cell death mechanisms can contribute to tumour development and progression [5], [6]. Among these mechanisms, apoptosis is a highly regulated form of cell death that plays an important role in maintaining cellular homeostasis and eliminating damaged cells; its intrinsic pathway is closely associated with mitochondrial function, linking apoptosis to mitochondrial processes such as oxidative phosphorylation and the activity of mitochondrial respiratory complexes [5]. Disruption of apoptotic signalling can contribute to uncontrolled cell survival.

This thesis investigates transcriptomic alterations in glioblastoma using data from the CPTAC GBM Discovery Study. Differential expression analysis is performed to identify genes whose expression differs between GBM and healthy brain tissue. Gene Ontology enrichment analysis is then used to characterise the biological processes associated with the significant genes, with particular attention given to cell-death-related processes. As apoptosis represents a prominent category among the identified cell-death-related processes, apoptosis-associated genes are subsequently investigated in relation to the mitochondrial respiratory complexes. Protein–protein association network analysis using STRING is used to examine associations between apoptosis-associated proteins and proteins belonging to the mitochondrial respiratory complexes.

The analysis focuses on the Classical molecular subtype of GBM, enabling the relationship between apoptosis and mitochondrial respiratory complexes to be examined within a defined transcriptional context. The aim of this thesis is therefore to characterise transcriptomic alterations in glioblastoma and investigate the transcriptional and network-level associations between apoptosis and mitochondrial respiratory complexes in Classical GBM through differential expression, functional enrichment, and protein–protein association network analysis.

Chapter 2 presents the relevant theoretical background, including glioblastoma, its molecular characteristics, apoptosis, mitochondrial function and respiratory complexes, and computational approaches for transcriptomic analysis. Chapter 3 describes the processing and analysis of the expression data, enrichment of the resulting significant gene set, and integration of cell-death- and mitochondrial respiratory-complex-related genes. Results from the analyses are presented in Chapter 4. An in-depth discussion of the results, limitations, and possible future work is presented in Chapter 5.

2 Theoretical Background

2.1 Glioblastoma

Glioblastoma multiforme (GBM) is one of the most aggressive and most common form of gliomas, with an extremely poor prognosis of a median survival period of 12-15 months [1], [7]. To date, GBM has no clear carcinogenetic cause, with high exposure to ionizing radiation being the only confirmed risk. Common carcinogenetic causes of other cancers such as smoking, diet, pesticide exposure, etc. have not been shown to cause GBM [1].

The World Health Organization classifies gliomas into grades I to IV, with grade I being relatively unproliferative and can be eliminated via surgery, and grade IV being the most invasive and highly malignant — GBM is classified under grade IV [1]. Additionally, GBM possesses an extreme heterogeneity in its structure which results in its resistance to conventional therapies and high recurrence rate [8]. A recent review article by David Eisenbarth and Y. Alan Wang [9] identified at least 10 major areas of heterogeneity in GBM's biology, presenting vast amounts of fronts to tackle therapeutically.

2.2 Molecular mechanisms of glioblastoma

2.2.1 Molecular heterogeneity and subtypes

GBM possesses an incredible molecular heterogeneity. Defining transcriptomic mutations of GBM include [8], [10]:

- ***TP53* mutations**: Appear in 31-38% of all GBM patients, play a role in disrupting cell cycle regulation and apoptosis.
- ***PTEN* mutations**: Appears in 24-37% of GBM patients, activate tumor-survival-enducing pathways.
- ***EGFR* amplifcation and mutations**: Appears in 36-60% of GBM patients, enhances cell growth.
- **and many others**.

The expression levels of the genes listed vary vastly on a per-sample basis [1], [2], [9]. This shows that GBM is not a single uniform disease, and different molecular profiles can correspond to different biological characteristics of the same disease.

According to Verhaak et al. and other further sequencing experiments [2], [9], GBM is categories into 3 major subtypes according to their expression of signature genes: Proneural, Mesenchymal, and Classical.

Each subtype have their own distinct levels of expression of the the defining genes and resistance to therapies. However, GBM samples have been shown to convert from one subtype to another during tumor progression and recurrence, contributing to its agressiveness and resistance to therapies. [9].

The Proneural subtype is characterized by major alteration of the PDGFRA gene and point mutation of *IDH1*, and it also had the most *TP53* mutations in the TCGA dataset; functional analysis points to developmental neuron-related processes. Mesenchymal subtype has genes in the tumor necrosis factor super family such as NF1 highly expressed, reflecting its tendency to have higher overall necrosis compared to other subtypes [2], [8].

According to Verhaak et al. [2], while *EGFR* are likely important across all subtypes, analysis shows that it has an even more impactful role in the Classical subtype where high-level *EGFR* amplification was observed to occur in 97% of the samples and infrequently occuring in other subtypes. Alongside that, there is a distinct lack of *TP53* mutation in the Classical subtype even though it is the most frequently mutated gene in GBM as a whole. Homozygous deletion of chromosome 9p21.3 also occured in 94% of the samples, affecting *CDKN2A*. This shows that the Classical subtype is characterized by a distinct molecular profile with alterations in regulatory pathways involving receptor signalling, cell-cycle regulation, and tumor suppression, justifying further investigation of this specific subtype [8]. The functional consequences of these alterations are discussed in the following section.

2.2.2 Major molecular pathways altered in GBM

Molecular alterations in GBM described above significantly affect pathways governing fundamental biological cellular processes, and the dysregulation of these processes promotes growth, invasion, and resistance to therapy in GBM [8].

A few of the most common pathways dysregulated (out of the several dozens) are [8], [10], [11]:

- **RTK/EGFR Signaling**: EGFR is a receptor tyrosine kinase whose activation stimulates downstream signalling pathways including PI3K/AKT/mTOR. Amplification or other activating alterations of EGFR can therefore result in persistent signalling that promotes tumour proliferation and survival.
- **PI3K/AKT/mTOR**: A central pathway influencing cell survival and proliferation with *EGFR* as a major upstream activator, the activation of which promotes tumor growth and treatment resistance. The tumor-supressing gene *PTEN* negatively affects the pathway, meaning the loss of it exacerbates GBM's aggressiveness.
- **p53**: *TP53* being a major component, the pathway governs cell cycle and apoptosis. Dysregulation leads to uncontrolled proliferation and impaired cell death.

Collectively, these alterations have the tendency to disrupt signalling pathways involved in cell proliferation, cell-cycle regulation, and cellular survival. Particularly, alterations affecting survival pathways allow abnormal genes to continue multiplying despite cellular damage or other signals that would otherwise signal their death, contributing to GBM's invasiveness [5]. As such, the dysregulation of cellular survival and death mechanisms are an important aspect to be inspected further. Among these mechanisms, apoptosis is particularly relevant because it is among one of the most common forms of regulated cell death [5].

2.3 Apoptosis in glioblastoma

Apoptosis, when functioning normally, is essential for normal development and tissue homeostasis, while its dysregulation has been implicated in various diseases, including cancer. In cancer, impaired apoptotic signalling can allow cells carrying genomic abnormalities to evade elimination and continue proliferating, thereby contributing to tumour development and progression. Consequently, the ability of cancer cells to evade apoptosis is considered an important characteristic of tumorigenesis [5], [11].

Apoptosis has 2 main pathways depending on the molecular interactions involved: (1) **intrinsic or mitochondrial pathway**, triggered by a variety of both extra- and intra-celullar stress, and (2) **extrinsic or death receptor pathway**, activated when extracellular ligands attach to certain extracellular receptors and resulting in the formation of death-inducing signalling complexes [5], [12]. More specifically, the cascades of interactions leading to apoptosis are [12]:

- **Intrinsic pathway**: stress induces the production of proteins in the Bcl-2 family which when inserted into the mitochondrial membrane, releasing cytochrome c. Cytochrome c interacts with Apaf-1 and procaspase-9, producing the apoptosome complex that triggers the activation of caspase-9,
- **Extrinsic pathway**: extracellular ligands include TNF (tumor necrosis factor), Fas-L (Fas ligand), and TRAIL (TNF-related apoptosis-inducing ligand), when attached to their respective extracellular portion of transmembrane receptors, forms a death-inducing signalling complex (DISC) and leads to the activation of caspase-8,

with both ultimately converging on the activation of executioner caspases which start the main process of apoptosis, better visualized in Figure 2.1. The main process involves loss of inner mitochondrial membrane potential, hyperproduction of superoxide ions, outflow of matrix calcium glutathione, and release of membrane proteins, rendering the cell “dead” [12].

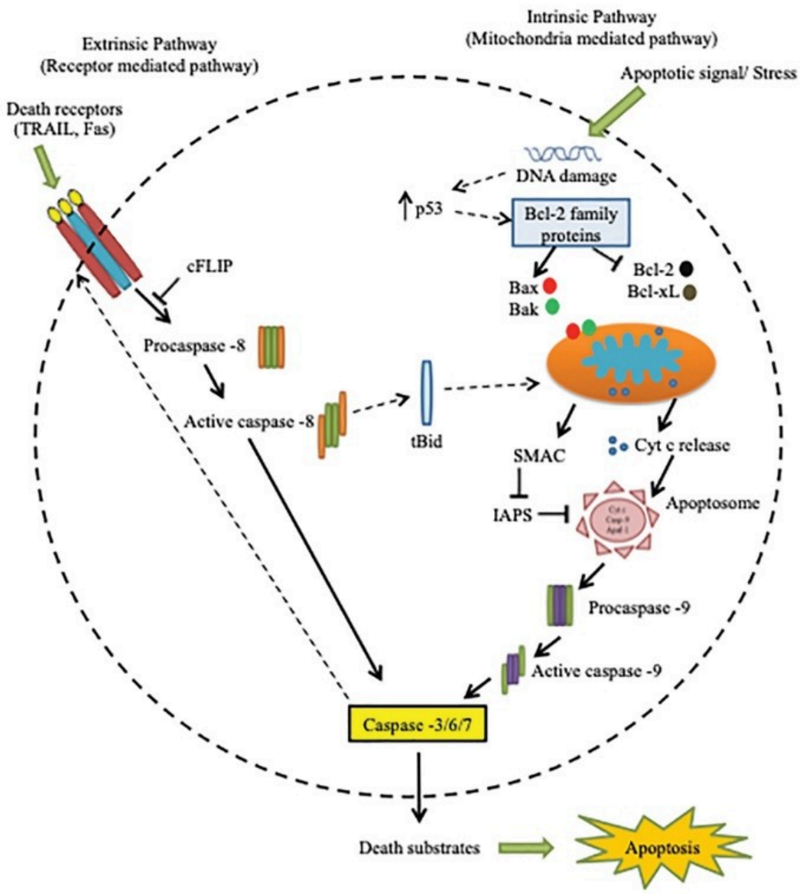

Figure 2.1. Pathways of Apoptosis [12]

The major molecular pathways affected in GBM discussed in the previous section, in one way or another, lead to the alteration of the apoptosis process described above. Altered apoptotic signalling can contribute to GBM's survival and resistance against multiple forms of therapy.

The intrinsic apoptotic pathway is closely associated with mitochondrial function. Mitochondrial outer membrane permeabilization and the subsequent release of cytochrome c represent key events in this pathway. At the same time, mitochondria are responsible for essential metabolic processes, including oxidative phosphorylation through the mitochondrial respiratory chain [5], [12]. This connection between mitochondrial function and apoptosis provides a basis for examining the relationship between apoptosis-associated genes and mitochondrial respiratory complexes, which is discussed further in the next section.

2.4 MITOCHONDRIAL FUNCTION AND RESPIRATORY COMPLEXES

2.4.1 GENERAL MITOCHONDRIAL FUNCTION

The mitochondria is the primary site of cellular function, and it has fundamental roles in processes such as ATP production, ROS generation, and the execution of cell death pathways [12], [13]. Carbohydrates, fatty acids, and amino acids can be metabolised into substrates, which enters the tricarboxylic acid (TCA) cycle. The TCA cycle generates reducing equivalents, primarily NADH and $FADH_2$, which subsequently donate electrons to the mitochondrial respiratory chain to drive oxidative phosphorylation and ATP production [13].

2.4.2 OXIDATIVE PHOSPHORYLATION AND THE RESPIRATORY CHAIN

Mitochondrial oxidative phosphorylation (OXPHOS) is a key mechanism of energy production in eukaryotic cells. It is comprised of 5 inner mitochondrial protein complexes I-V (also referred to as CI, CII, etc.) and 2 mobile electron carriers forming the electron transport chain (ETC) [14].

The ETC **oxidizes** reducing equivalents in NADH and succinate, inducing the reduction of molecular oxygen to water and the pumping of protons across the inner mitochondrial membrane (IMM) via Complex I, III, and IV, each using their own pumping mechanism, to drive ATP synthesis [14]. The ETC also uses energy from the reducing equivalents to generated a proton gradient across the IMM, which is used by ATP synthase (Complex V) to **phosphorylate** ADP to ATP, the universal energy currency of cells . The specific roles of each complexes are as follows [14]:

- **Complex I**: Transfers electrons from NADH to ubiquinone and pumps protons.
- **Complex II**: Does not pump protons but contributes to reduction of ubiquinone.
- **Complex III**: Reduces ubiquinone and cytochrome c, in turn shuttles to Complex IV.
- **Complex IV**: Donates electron for final reduction of oxygen.
- **Complex V**: Uses proton gradient and the reduction of oxygen to synthesize ATP.

The activity of the respiratory chain is therefore essential for maintaining mitochondrial energy production and cellular function.

2.4.3 MITOCHONDRIAL RESPIRATORY CHAIN-APOPTOSIS RELATIONSHIP IN GBM

The mitochondrial respiratory chain and (intrinsic) apoptosis can be seen to be inherently closely connected to each other. While the respiratory chain is essential for maintaining mitochondrial energy production and homeostasis, intrinsic apoptosis depends on the mitochondria for the regulation of apoptotic signalling. Therefore, disruption of respiratory-chain function and mitochondrial homeostasis can influence apoptotic signalling particularly through cytochrome c because of its involvement in both cellular respiration and apoptosis (Figure 2.2) [15], [16].

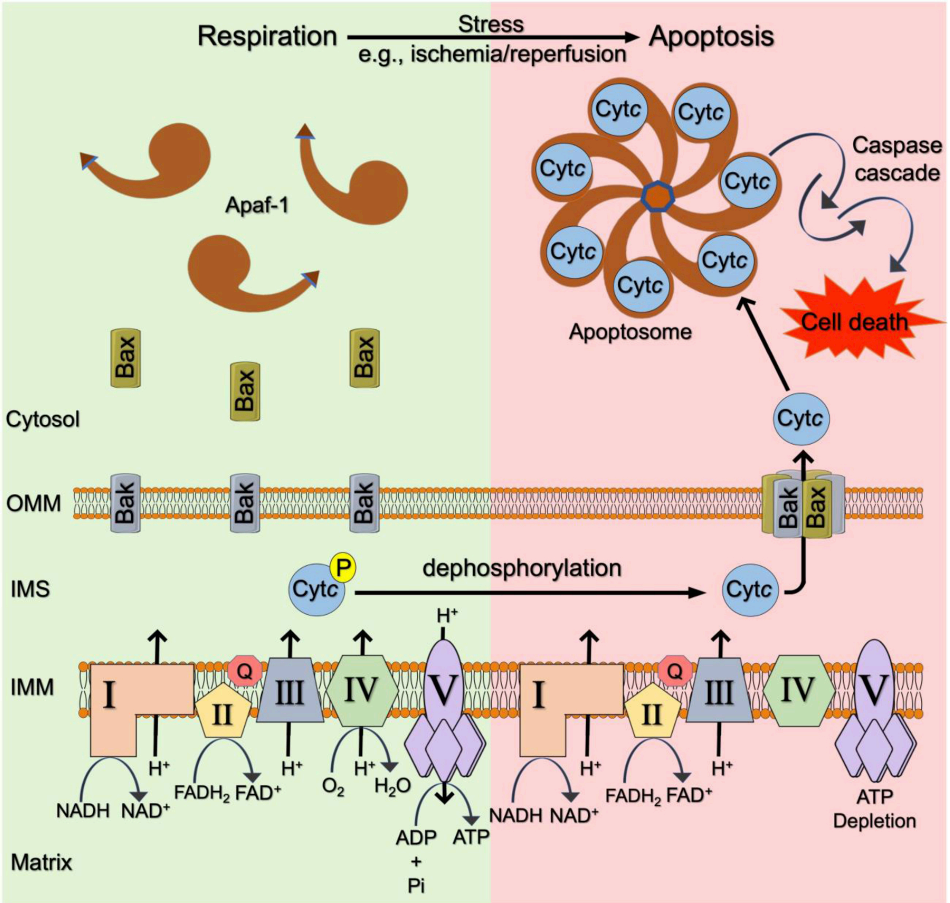

Figure 2.2. Proposed cytochrome c involvement in cellular respiration and apoptosis [16]. Cytochrome c acts as an electron carrier between Complex III and IV. Cellular stress causes cytochrome c to be released from the mitochondria, followed by a variety of interactions that produce apoptosome, resulting in apoptosis [12], [16].

This relationship is particularly relevant in GBM, where alterations in mitochondrial metabolism and oxidative phosphorylation have been implicated in tumour-cell survival and progression in specific subtypes [15], [17]. Changes in respiratory-chain activity may therefore influence the susceptibility of GBM cells to apoptosis. More specifically, improper functioning of the mitochondria may result in improper apoptosis, resulting in promoted tumor cell survivability.

2.5 Transcriptomics and computational analysis

2.5.1 From conventional molecular analysis to computational approaches

Having established the relevance of apoptosis and mitochondrial respiratory-chain function in GBM, investigating alterations in the genes involved in these processes requires methods capable of measuring gene activity across large numbers of genes simultaneously.

Understanding complex biological systems requires integration of both experimental and computational approaches [18]. Kitano [19] argues that although studying individual components of an organism can provide valuable insights, such approaches alone are insufficient to understand the behaviour of the system as a whole. In molecular pathology, diseases are rarely attributable to a single gene, but instead arise from complex interactions between multiple molecular components and biological processes [20].

Although systems-level approaches to biology had already gained considerable interest, their application was historically constrained by limitations in experimental technologies and the availability of quantitative molecular data [19]. Recent advancements has made high-throughput and high quality profiling of biological systems possible, bringing forth a new set of challenges and opportunities in computationally analyzing and interpreting the new massive amount of deposited data, particularly with DNA microarrays and Mass Spectrometry (MS)-based proteomics [21], [22].

Among these approaches, transcriptomic technologies provide a means of measuring genome-wide changes in gene expression, which can subsequently be analysed computationally to identify molecular alterations associated with disease. For example, transcriptomic profiling has enabled the identification of distinct molecular subtypes of GBM, presenting more specific areas of deeper research into the disease [2].

2.5.2 Transcriptomics and differential expression analysis

Transcriptomics technologies provide information on gene expression by measuring the RNA transcripts produced from an organism's genome [3]. According to Lowe et al. [3], two major approaches for measuring transcript abundance are microarrays, which quantify predetermined sequences, and RNA sequencing (RNA-seq), which uses high-throughput sequencing to measure transcript sequences without requiring them to be predetermined.

Large-scale efforts have been made to profile the transcriptomes of cancers, including GBM. Public resources such as the Gene Expression Omnibus (GEO) and The Cancer Genome Atlas (TCGA) provide access to extensive molecular profiling datasets from numerous cancer types, including glioblastoma [23], [24]. The Clinical Proteomic Tumor

Analysis Consortium (CPTAC) has further generated and integrated transcriptomic and proteomic data from cancer studies, including GBM [25]. These resources provide opportunities for researchers to computationally reanalyse previously generated data and investigate molecular characteristics of cancer. The next challenge is therefore to extract meaningful biological information from these large-scale quantitative datasets.

Comparative profiling involves comparing molecular measurements between predefined biological conditions to identify differences associated with a particular phenotype or state [26]. In transcriptomics, differential expression analysis is commonly used to compare gene expression between conditions, such as diseased and healthy tissue. Its main objective is to identify genes whose expression differs between the conditions, providing insight into potential molecular mechanisms underlying the observed phenotype.The resulting differential expression statistics can describe both the magnitude and direction of an expression difference, commonly represented by the log2 fold change (log2FC), and its statistical significance, represented by a p-value. Thresholds can subsequently be applied to identify genes considered differentially expressed, with the choice of thresholds depending on the experimental context and research objectives [27].

A popular library for gene-level differential expression analysis is DESeq2 (with PyDESeq2 providing a Python implementation). An average workflow with DESeq2 is displayed in Figure 2.3.

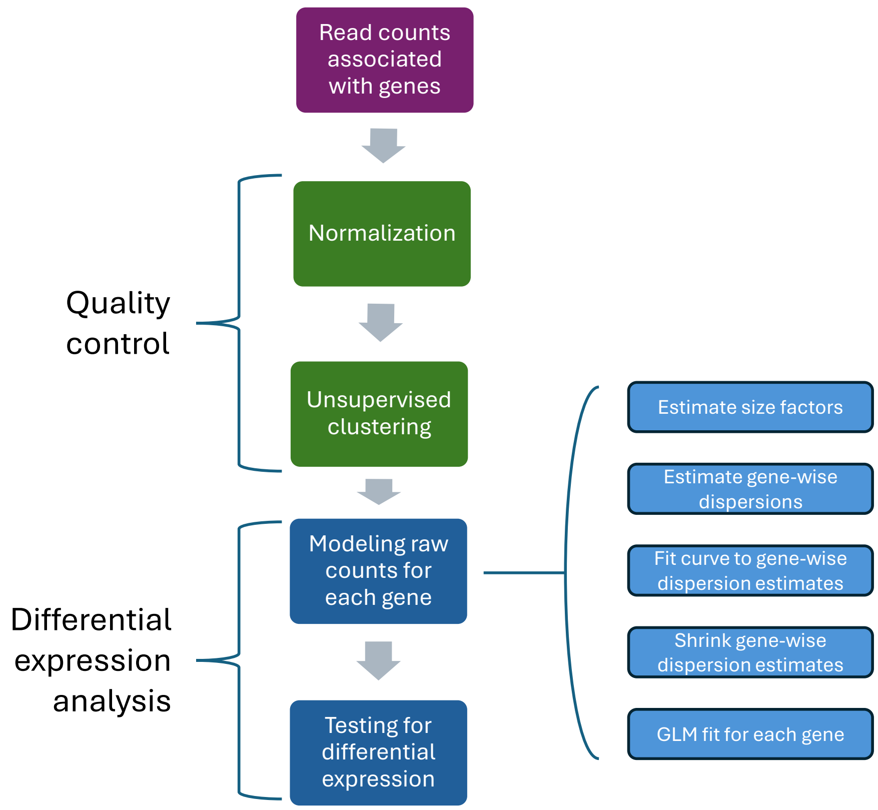

Figure 2.3. Conceptual workflow of differential expression analysis using DESeq2. Adapted from Harvard Chan Bioinformatics Core (HBC) [28].

RNA-seq count data often exhibit greater variance than their mean, a property known as overdispersion. DESeq2 accounts for this by modelling count data using a negative binomial generalized linear model, from which $\log_2$ fold changes and their standard errors are estimated. The statistical significance of the estimated $\log_2$ fold changes is then assessed using Wald tests. DESeq2 can additionally account for known sources of systematic variation, such as batch effects, by incorporating relevant variables into the design of the statistical model. To account for the increased number of false discoveries resulting from multiple hypothesis testing, p-values are adjusted using the Benjamini–Hochberg procedure to control the false discovery rate (FDR) [29], [30], [31].

2.5.3 FUNCTIONAL ANALYSIS AND NETWORK ANALYSIS

Interpreting differentially expressed genes (DEGs) individually may provide limited insight into the broader biological processes underlying the observed changes [19]. Functional enrichment analysis can instead identify biological pathways or functions that are over-represented among the identified DEGs. [32]

The Gene Ontology (GO) database provides structured and standardized representation of biological activities according to gene products, organized into three main aspects: (1)

Molecular Functions (MF), (2) Cellular Components (CC), and (3) Biological Processes (BP) [33]. GO enrichment analysis evaluates whether particular GO terms are statistically overrepresented among a set of genes compared with an appropriate background gene set (Figure 2.4) [32], [34]. The process allows groups of DEGs to be interpreted in terms of shared biological functions and processes rather than as individual genes.

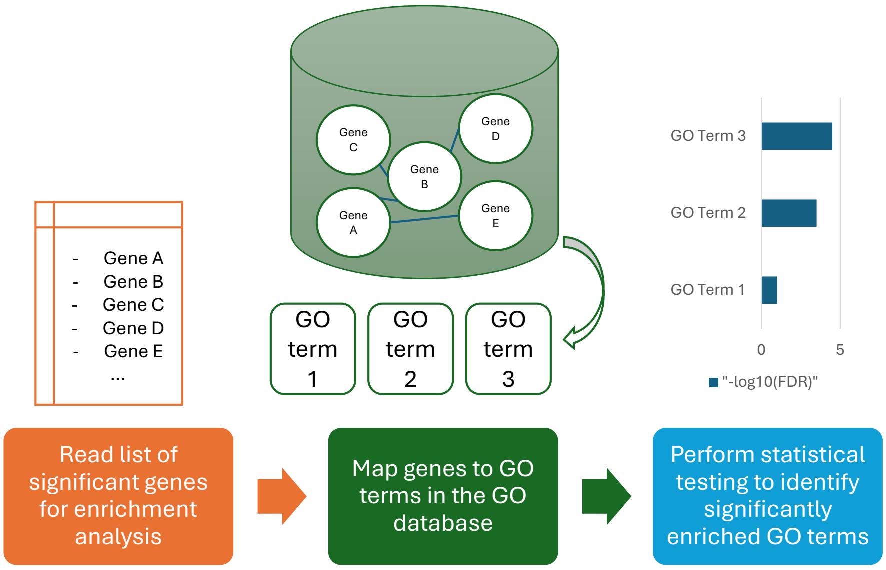

Figure 2.4. Workflow of GO enrichment analysis. Adapted from The Gene Ontology Consortium [35].

Still, the identified overrepresented biological processes do not show how the individual genes are interacting with one another. Therefore, a network-based approach can complement the identified BPs by representing the genes as nodes and their interactions as edges, allowing for the inspection at the systems-level [20].

Network-based approaches can reveal highly connected components, clusters of interacting molecules, and relationships between biological processes that may not be apparent when genes are considered individually. This can provide additional context for interpreting molecular alterations and identifying potential interactions between genes involved in related biological processes [20].

In this thesis, Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) was used to construct protein–protein association networks from selected differentially expressed genes. STRING maps the submitted gene identifiers to their corresponding protein products and integrates known and predicted protein associations from multiple evidence sources. The resulting networks represent proteins as nodes and their associations as edges, providing a means of examining functional relationships among the proteins encoded by the selected DEGs [36].

Together, functional enrichment and network analysis therefore enable differentially expressed genes to be interpreted at a systems level, providing both information on the biological processes associated with gene-expression changes and the molecular relationships connecting the genes involved.

3 Materials and Methods

3.1 Data Source

The transcriptomic data were retrieved from the National Cancer Institute's publicly available Clinical Proteomic Tumor Analysis Consortium Glioblastoma Multiforme (CPTAC-GBM) cohort through the NCI's Genomic Data Commons (GDC) Data Portal. The GDC dataset contained RNA-seq data from over 200 GBM samples in the form of raw counts of each gene. CPTAC-GBM Discovery Study metadata were then used to identify the molecular subtypes of the corresponding cases. Among the CPTAC cases, 99 samples with the Proneural, Mesenchymal, and Classical molecular subtypes were identified and used for the transcriptomic analysis [37], [38]. Of the 99 samples, 25 were identified to be of the Classical subtype.

Additionally, the CPTAC-GBM Discovery Study used 10 normal frontal cortex samples from the Genotype-Tissue Expression (GTEx) project as controls. Of these 10 GTEx controls, transcriptomic data for 7 were accessible and were therefore used as controls in this thesis [38]. The same GTEx controls were selected to maintain consistency with the original CPTAC-GBM study and to facilitate potential future comparisons between transcriptomic and proteomic data. Differences in sequencing depth and library size between samples were accounted for through the size-factor normalization performed by PyDESeq2 during differential expression analysis [30], [31].

3.2 Data quality control and inspection

As suggested by the Harvard Chan Bioinformatics Core [39], with raw gene counts, a number of plots can be used to inspect the general landscape of the data and assess its quality, mainly through hierarchical clustering methods and Principal Component Analysis (PCA) of log2-transformed count data. Because transcriptomic expression values can span several orders of magnitude, log2 transformation compresses the range of expression values and reduces the influence of highly expressed genes, thereby improving the suitability of the data for clustering and visualization [39], [40], [41].

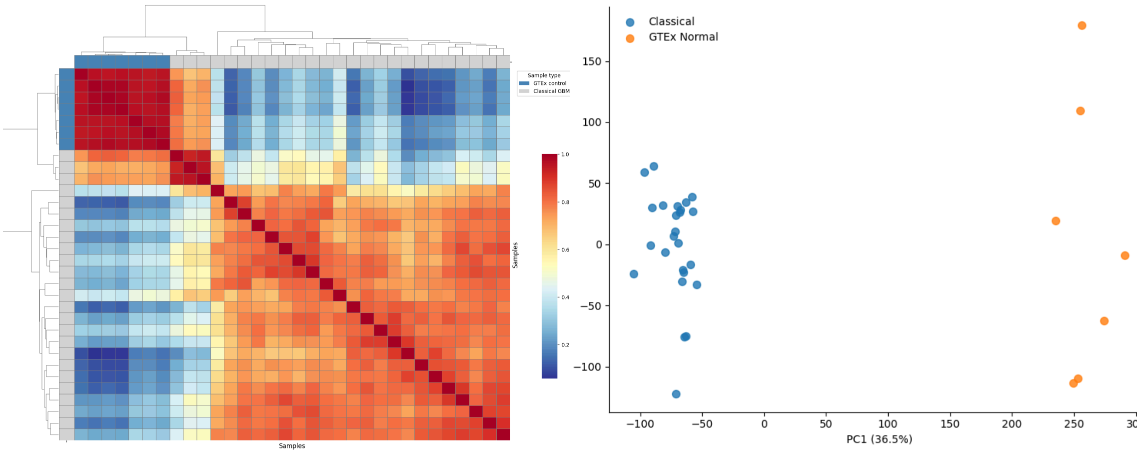

Figure 3.1. Hierarchical Clustering Heatmap (left), PCA (right)

The hierarchical clustering heatmap (shown in Figure 3.1) groups samples according to their similarity based on the samples' gene expression Pearson correlation. Samples with similar biological characteristics are expected to exhibit similar expression profiles and therefore cluster together [39]. In the current dataset, most Classical GBM samples form a distinct cluster from the control samples. However, three samples cluster more closely with the controls than with the other GBM samples. To determine whether these samples represent potential outliers or reflect biological variation, PCA is a complementary quality control plot used for further data quality confirmation.

As noted by the Harvard Chan Bioinformatics Core [39], biological replicates are expected to have similar expression profiles and cluster together in PCA. The PCA (also shown in Figure 3.1) demonstrates clear separation between Classical GBM and control samples, with no apparent sample-level outliers. Thus, although three samples show higher similarity to the controls in the hierarchical clustering analysis, their position in the PCA does not indicate clear evidence of them being outliers.

3.3 DIFFERENTIAL EXPRESSION ANALYSIS

The raw count data from the 25 Classical GBM samples and 7 GTEx control samples were used as input to the PyDESeq2 model. PyDESeq2 performed differential expression analysis and generated the corresponding log2FC, standard error, test statistic, p-value, and adjusted p-value for each gene. PyDESeq2 models RNA-seq count data using a negative binomial generalized linear model and assesses the statistical significance of the estimated coefficients using Wald tests [30], [31].

With the log2FCs and their adjusted p-values, volcano plots were used to display the distribution of differentially expressed genes (DEGs). DEGs $|\log_2 \mathrm{FC}| > 1$ and adjusted p-value < 0.05 were considered significantly differentially expressed. These thresholds were selected to identify genes showing both a minimum two-fold change in expression and statistically significant differential expression. The volcano plot along with the count of significant DEGs are presented in 4.1.

3.4 GO FUNCTIONAL ENRICHMENT

GO enrichment analysis was performed on the set of significant differentially expressed genes (DEGs) to identify overrepresented Biological Process (BP) terms. The analysis was performed using the GSEApy Python library with the predefined `GO_Biological_Process_2023` gene-set library. The significant DEGs identified from the differential expression analysis were used as the input gene set. The resulting enrichment terms were ranked according to their statistical significance and the number of genes involved each of the enriched term, and displayed in a dot plot. The results are presented in 4.2.

The top enriched biological processes were subsequently examined to identify processes relevant to the objectives of this study. Based on the enrichment results and discussed biological connection in 2.4.3, apoptosis and mitochondrial respiratory-chain complexes were selected for further analysis.

3.5 STRING INTERACTION NETWORK ANALYSIS

The list of genes associated with apoptosis was retrieved from the Kyoto Encyclopedia of Genes and Genomes (KEGG) database, via GSEApy using the `KEGG_2021_Human` gene-set library [42]. The list of genes complexes I–V was retrieved from Human Mitocarta [43]. DEGs belonging to either the apoptosis gene set or the mitochondrial respiratory-complex gene sets were combined into a single gene set for further analysis.

The selected genes are then fed into to the STRING database to obtain protein association networks. The resulting networks were imported into Python and analysed using the NetworkX library. Associations between apoptosis-associated proteins and proteins belonging to each mitochondrial respiratory-chain complex were identified by counting the corresponding network edges. The resulting edge counts were used to characterize the associations between apoptosis and complexes I–V.

4 Results

4.1 Differentially Expressed Genes in GBM

Presented in Figure 4.1 are plots plotting the transcriptomic log2FC against proteomic log2FC.

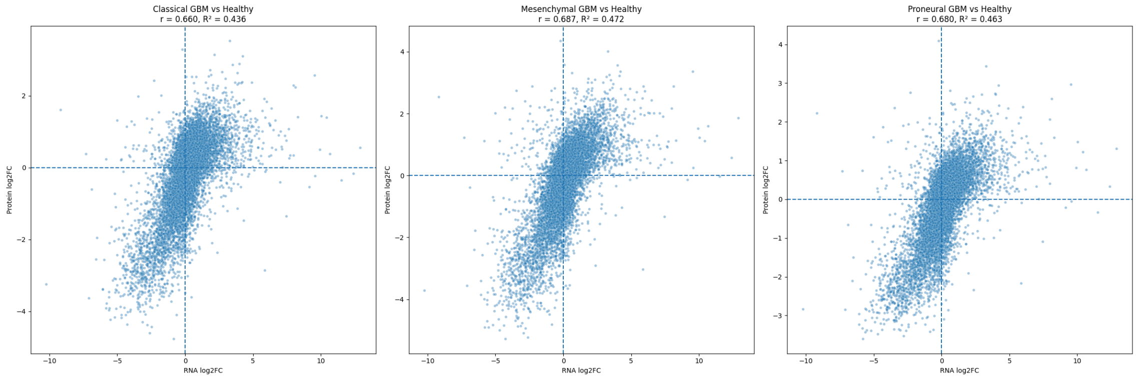

Figure 4.1. Log2FC comparison of transcriptome and proteome for classical (left), mesenchymal (middle), proneural (right) subtypes.

For classical, mesenchymal, and proneural subtypes respectively, the calculated squared-Pearson correlation values R^2 are 0.436, 0.472, and 0.463. The results line up quite closely with a previous major study conducted by Vogel and Marcotte [44], stating that only about 40% of the variation of one can be explained by the other.

4.2 Functional enrichment of differentially expressed genes

4.3 Association between apoptosis and mitochondrial respiratory complexes

5 DISCUSSION

5.1 CYTOCHROME C INVOLVEMENT IN APOPTOSIS AND CELLULAR RESPIRATION

5.2 LIMITATIONS AND FUTURE IMPROVEMENTS

6 CONCLUSIONS

7 INTRODUCTION

This document template conforms to the Guide to Writing a Thesis in Tampere University [45], [46]. A thesis typically includes at least the following parts:

- Title page
- Abstract in English (and in Finnish)
- Preface
- List of abbreviations and symbols
- (Lists of figures and tables)
- Contents
- Introduction
- Theoretical background
- Research methodology and material
- Results and analysis (possibly in separate chapters)
- Conclusions
- References
- (Appendices)

Each of these is written as a new chapter, in the files in the folders `frontmatter/`, `mainmatter/` and `appendices`, which are then included into the `index.typ` files in said folders using the Typst directive `#include`. Read this document template and its comments carefully. The titles of Chapters from 1 to 10 are provided as examples only. You should use more descriptive ones. The title page and abstract headers are created by inserting the relevant information into the file `metadata.typ`. The table of contents lists all the numbered headings after it, but not the preceding ones.

Introduction outlines the purpose and objectives of the presented research. The background information, utilized methods and source material are presented next at a level that is necessary to understand the rest of the text. Then comes the discussion regarding the achieved results, their significance, error sources, deviations from the expected results, and the reliability of your research. The conclusions form the most important chapter. It does not repeat the details already presented, but summarizes them and analyzes their consequences. A list of references enables your reader to find the cited sources.

An introduction ends in a paragraph, that describes what each following chapter contains. This document is structured as follows: Chapter 8 discusses briefly the basics of writing and presentation style regarding the text, figures, tables and mathematical notations. Chapter 9 summarizes the referencing basics. Chapter 10 presents a discussion on the discoveries made during this study.

8 Writing practices

Effective written communication requires both sound content and clear style. Keep the layout of your thesis neat and pay attention to your writing style.

8.1 Text

Do not worry about the layout of the text, this template takes care of it already. Brief basics of writing style:

- Always think of your reader when you are writing and proceed logically from general to specific.
- Highlight your key points, for example, by discussing them in a separate section or subsection, or presenting them in a table or figure. Use *italics* (`_text_`) for emphasis, but don't overdo it.
- Avoid long sentences and complicated statements. A full stop is the best way to end a sentence.
- Use active verbs to make a dynamic impression but avoid the first person pronoun I, except in your preface.
- Avoid jargon and wordiness. Use established terminology and neutral language.
- The minimum length of sections and subsections is two paragraphs, and you need to consider the balance of chapters. Paragraphs must always consist of more than one sentence.
- Do not use more than three levels of numbered headings, such as $4.4.2$. However, keep in mind that 3 different levels of sections is already a bit much.
- Do not use too many abbreviations. Use capital and small letters consistently. You might wish to define new commands in the `preamble.typ` file, for words you do not wish to misspell. For example, the `preamble.typ` file defines the command `#eeg`, which is typeset as electroencephalography.

Your text will also look better if you do not extensively rely on lists such as the above one. Ideally, a list should not end a section either, which is why this paragraph is here.

8.2 Images

You must refer to all the images in the body text. The reference should preferably appear on the same page as the actual image or before it. Images and tables must be numbered consistently and primarily placed at the top of the page, but you are free to decide where they fit best.

8.2.1 IMAGES IN TYPST

Typst takes care of the numbering, if you specify a unique label `<label-name>` right after the a image or a table. Cross-referencing is done by prefixing the identifier inside the angle brackets with a `@` symbol: `@label-name`. For example, the code

```
#figure(
    image(
        "../images/tau-logo-fin-eng.svg",
        alt: "Tampere University logo with a human face formed from geometric shapes on the left and the university name on the right.",
        width: 80%
    ),
    caption: [Tampere University logo.],
) <tau-logo>
```

is rendered as seen in Figure 8.1.

Tampereen yliopisto
Tampere University

Figure 8.1. Tampere University logo. [47]

To showcase how figures with subfigures are generated, Figure 8.2 displays a few different gradient or color progression shapes. Other colormaps than the `rainbow` seen here are also available. The subfigures are generated by calling the function `tauthesis.subfigure` inside of a figure of type `image`.

(a) (b) (c)

Figure 8.2. A display of a few different color gradient shapes available in Typst. These include the linear gradient in Figure 8.2.a, the radial gradient in Figure 8.2.b and the conic gradient in Figure 8.2.c.

Never start or end a chapter with a image, table, equation or list. The caption is placed under an image. The contents of all images must be explained in the text body, so that the readers know what they are supposed to notice. Images generated by analysis software usually need further editing. The figures should be in the same language as other text. The recommended font size is the same as that of the body text, 11 pt. All images must be readable, even if your thesis is printed in greyscale. Whenever possible, use images in vector formats such as `.svg`, as they can be scaled without loss of quality. Also note that as of Typst version 0.14.0, including PDF files as images should work natively. However, see the below accessibility considerations for why this might not work for your thesis.

8.2.2 ACCESSIBILITY OF IMAGES

All images produced via the `image` [48] function (or otherwise) *must* contain so-called *alternative texts* [49], provided via the named argument `alt` . These **must** be provided for your thesis to remain accessible. Even if an `image` is inside of a captioned `figure` [50], an alternative text is still needed. The alternative text also needs to be more descriptive than the figure caption, because an alternative text is the only indication of what goes on in an image for people relying on assistive technologies. You can see an example of these principles being applied in Figure 8.1.

Also note that since most PDF files produced programmatically do not contain tags, and Typst has no way of automatically deducing what the tags should have been like, including PDF images does not work with the accessible PDF standards such as PDF/UA-1, PDF/A-3a and others. You should therefore prefer SVG images when inserting vector graphics into your thesis.

8.3 TABLES

Tables are convenient for presenting information in a concise way, especially numerical data. Tables have numbered captions, see Table 8.1 for an example. The caption is placed on the same page but above the table, unlike the captions that accompany images. You must refer to all the tables in the body text. In addition, you must discuss the content of any tables in the body text to ensure that readers understand their relevance.

Mark the titles of the columns and units clearly. You can use underscores `_emph_` to highlight the titles, if necessary. The order of the columns and rows must be carefully considered. Ideally, all cells should *not* be surrounded with a border, as it may make your table harder to read. The numbers should be aligned at the decimal point for easy comparison. You should preferably use SI units, established prefixes and rewrite large numbers so that the power of ten should be placed in the title of the column instead of each row, if possible. More suggestions can be found in [51].

8.3.1 TABLES IN TYPST

To actually draw a table that can be referenced, we issue a `#figure` call with a `table` as its argument:

```
#import table: cell, header, hline, vline
#[
    #show table.cell.where(y: 0): strong
    #figure(
        table(
            columns: 5,
            stroke: none,
            align: center + horizon,
            table.header(
                [Substance],
                table.vline(),
                [Pressure (#unit(pascal))],
                table.vline(),
                [Pressure (#unit(bar))],
                table.vline(),
                [Pressure (#unit(mmHg))],
                table.vline(),
                [Temprature (#celsius)]
            ),
            table.hline(),
            [Mercury],
            num(1.0),
            num(0.0001),
            num(0.0075),
            num(41.85),
            [Tungsten],
            num(1.0),
            num(0.0001),
            num(0.0075),
            num(3203),
        ),
        caption: [Vapour pressures of two metals. @wikipedia-vapor-pressure],
    ) <tab-evaporation-conditions>
]
```

The added `#[...]` around the whole figure restricts the scope of the show rule that boldens the header to this table. This results in Table 8.1 being displayed.

Table 8.1. Vapour pressures of two metals. [52]

Substance	**Pressure** (Pa)	**Pressure** (bar)	**Pressure** (mmHg)	**Temprature** (°C)
Mercury	1	0.000 1	0.007 5	41.85
Tungsten	1	0.000 1	0.007 5	3 203

A table should not end a section. That is why after every table we state some observations about the table. Here we notice that mercury and tungsten have identical vapour pressures, but at very different tempratures.

8.3.2 ACCESSIBILITY OF TABLES

Tables do not need alternative texts, not even if you export your document using the PDF/UA-1 standard. Screen readers will read the table in the order in which your table cells were given in Typst source code. It is also important to properly use the `table.header` and `table.footer` for accessibility tags to be applied properly.

The example table shows how to typeset a table in the usual *row-major* order, where the displayed order in the code is the same as it is in the final PDF file. There are situations, such as when a table takes too much space horizontally, where you might wish to *transpose* a table in the PDF output, so the output is in *column-major* order. Typst does not currently support this in tables that aim to be semantically correct, as table headers and footers constructed with `table.{header,footer}` are always typeset horizontally, and `table.cell` show rules currently (as of 2025-10-09) do not have access to the `x` and `y` positions of each cell.

Also note that accessible tables should be *regular* [53]: every row of a table should contain an equal number of columns or cells. In other words, table cells spanning multiple columns or rows are heavily discouraged. For the more mathematically oriented, it could be said that tables should only be used to represent *relations*. Table 8.1 is an example of such a regular table.

If your table is getting very wide, consider simply scaling it to the page width using the `scale` function. Note however that this will make the table contents tiny, and less accessible to people who do not see well, but still do not rely on screen readers. Rotating tables by $90°$ using the `rotate` function is also possible.

8.4 MATHEMATICAL NOTATIONS AND EQUATIONS

Numbers are generally written using numerals for the sake of clarity, for example “6 stages” rather than “six stages”, which is nevertheless strongly preferred to “a couple of stages”. You should also use a thousand separator, i.e. instead of 55700125 write $55\,700\,125$. Never omit the leading zero in decimals. A comma is used as a decimal separator in the Finnish language and a period in the English language. You might wish to use the Typst library **zero** for typesetting numbers, even if the file `preamble.typ` does provide a rudimentary function `num` for grouping number digits.

Like numbers, it is advisable to abbreviate units of measurement. There is a space between the number and the unit, but you must keep them on the same line. The space is somewhat shorter than a word space, see 1 μm and $1\,\mathrm{\mu m}$ for comparison. It is better to compile a table or graph than include a great deal of numerical values in the body text. Use precise language and put numbers on a scale (small, fast, expensive).

Use generally known and well defined concepts and standard conventions and symbols for representing them. New concepts should be defined when they appear in the text for the first time. Upper case and lower case letters mean different things in symbols and units of measurement. Do not use the same symbol to mean different things, but note that different styles applied over the same symbol are accepted: if $\mathbf{x}$ is a position vector, then the scalar $x = \|\mathbf{x}\|$ could be used to denote its norm. If the position is a random variable, then one could use yet another style such as fraktur form $\mathfrak{x}$ of $\mathbf{x}$ to eccentuate this fact.

8.4.1 MATHEMATICS IN TYPST

Strings of mathematical symbols such as $\Theta(n^2)$ are typeset in Typst using the math mode, between dollar signs: `$math$` for inline mathematics or

```
$
    math
$
```

with whitespace surrounding the contents for display math. Simple formulas may be displayed within the body of the text without numbering. As an example of a highlighted formula, the Fundamental theorem of calculus can be written in the following way:

$$\int_{[a,b]} f(x)\,\mathrm{d}x = F(b) - F(a)\,. \tag{8.1}$$

Here f is the *integrand*, F its *anti-derivative* and $[a, b]$ the integration interval. Please note that all the variables must be defined at the point of their first appearance. The formulae are shown in a different typeface on purpose and the symbols are almost always italicised. Vectors can be written in slanted or upright boldface, or with arrows:

$$\begin{aligned} \boldsymbol{v} &= \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} \\ \mathbf{v} &= \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \\ \vec{v} &= \frac{\mathrm{d}\vec{x}}{\mathrm{d}t} \end{aligned} \tag{8.2}$$

The most important thing is to be consistent throughout your thesis. Do not mix and match notations, as it will confuse the reader. See the file `preamble.typ` for an example of how you might define a `#vector` command for typesetting vectors consistently.

Defining a command also allows you to easily switch notations globally, if you are not satisfied with your current choice.

Units are written in an upright font with a normal weight, to distinquish them from variables:

$$\|\mathbf{f}\| = m\|\mathbf{a}\| = 10\,\mathrm{kg} \cdot 9.81\,\frac{\mathrm{m}}{\mathrm{s}^2} = 98.1\,\mathrm{N}\,. \tag{8.3}$$

Here $\mathbf{f}$ is a force vector, m is the mass of the object experiencing the force, and $\mathbf{a}$ is its acceleration vector. Again, see the `preamble.typ` file for how you might define commands for units.

Mathematical formulae are numbered, if they are written on separate lines and referred to in the main body of the text. The number is usually put in parenthesis and right aligned, see (8.1) for an example. If you wish to leave out the number from a highlighted equation, wrap it into the function `math.equation` with the named arguments `numbering: none` and `block: true`:

```
#math.equation(
    numbering: none,
    block: true,
    alt: "force = mass acceleration .",
    $force = mass acceleration thin .$
)
```

This results in the unnumbered block-level equation

$$\mathbf{f} = m\mathbf{a}\,.$$

Include any punctuation (commas, periods, …) surrounding an equation in the equations themselves, as is again shown in (8.1). The command `thin`, which in mathematics mode is a shorthand for `sym.space.thin`, can be used to insert just a little extra space between elements such as the math itself and punctuation, which is a good readability-increasing practice. The added visual space is not relevant for the alternative text of the equation, though, so `thin` should be removed from it, as is shown above.

Do not start a sentence with a mathematical symbol but add some word, such as the name or type of the symbol in front of it. Variables, such as x and y, are generally presented in italics, whereas elementary functions, special functions and multi-letter operators are not:

$$\sin(2x + y) \quad \text{or} \quad \lim_{x \to -1} \frac{x^2 - 1}{x + 1} = -2\,. \tag{8.4}$$

As a rule of thumb, it is best to rely on the automated formatting of an equation editor, such as the one provided by Typst.

Vectors and matrices are written with the `vec` and `mat` functions:

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{bmatrix} = \mathbf{b}\,. \tag{8.5}$$

See the Typst documentation on mathematics [54] for examples of how different symbols are written.

8.4.2 Accessibility of mathematics

If you have been reading the source code of the PDF file of this example template, you might have noticed that in addition to the math mode dollar signs `$math$` and

```
$
    math
$
```

the template has explicitly been using the Typst function `math.equation` [55] to include mathematics into a document. This is because the function allows one to provide an *alternative text* for each equation, as is required by the PDF accessibility standards. This is done via the alternative text argument `alt` of the function:

```
#math.equation(
    alt: "Alternative text here",
    $equation here$
)
```

You might have noticed that the alternative texts in the example mainly use the Typst source code as the alternative text representation. **Note that this is only acceptable, if your equation source code carries the semantics of the displayed equation with it.** Note the difference between the *completely unacceptable*

```
#math.equation(
    alt: "f = m a",
    $vector(f) = m vector(a)$
)
```

the slightly better option

```
#math.equation(
    alt: "vector(f) = m vector(a)",
    $vector(f) = m vector(a)$
)
```

and finally the most descriptive one, where the commands `force`, `mass` and `acceleration` have been defined in the file `preamble.typ`:

```
#math.equation(
    alt: "force = mass acceleration",
    $force = mass acceleration$
)
```

Remember that in equation (8.3) the text styles (bold versus normal weight) were used to differentiate between scalars and vectors. Force and acceleration are implicitly vectors, and mass is implicitly a scalar. Having the alternative text mention what the actual quantities are instead of using just letters makes the intention clearer. It also allows you to copy and paste the source code into the alternative text without any modifications.

You might also wish to use the function `altTextFn` provided in `preamble.typ` to clean up extra whitespace from your alt texts, if you wish to write them on multiple lines. Indentation and added space are not automatically trimmed from alt texts by Typst.

8.5 PROGRAMS AND ALGORITHMS

Codes and algorithms are written using a monospaced font. If the length of the code or algorithm is less than 10 lines and you do not refer to it later on in the text, you can present it similarly to formulas. If the code is longer but shorter than a page, you present like a figure titled “Listing” or “Algorithm”. In Typst, code blocks are presented in backticks:

````
```typst
#figure(
    raw(
        read("../code/square.jl"),
        lang: "julia",
        block : true
    ),
    caption: [
        The square of
        #math.equation(
            alt: "x in real numbers",
            $x in bb(R)$
        )
        and
        #math.equation(
            alt: "x in complex numbers",
            $x in bb(C)$
        ),
        written in Julia @Bezanson2017Julia.
    ],
) <program-example>
```
````

The above results in what is seen in Listing 8.1.

```
#figure(
    raw(
        read("../code/square.jl"),
        lang: "julia",
        block : true
    ),
    caption: [
        The square of
        #math.equation(
            alt: "x in real numbers",
            $x in bb(R)$
        )
        and
        #math.equation(
            alt: "x in complex numbers",
            $x in bb(C)$
        ),
        written in Julia @Bezanson2017Julia.
    ],
) <program-example>
```

Listing 8.1. A code block.

Removing the backticks would actually load code from the file `code/square.jl` to be displayed in the numbered Listing 8.2. The listing demonstrates how multiple dispatch allows you to define methods for functions with different argument types in the Julia programming language.

```
"""
  out = square(x::Real)

Computes the square of an 𝑥 ∈ ℝ.
"""
function square(x::Real)
  x * x
end

"""
  out = square(x::Complex)

Computes the square of an 𝑥 ∈ ℂ.
"""
function square(x::Complex)
  conj(x) * x
end
```

Listing 8.2. The square of $x \in \mathbb{R}$ and $x \in \mathbb{C}$, written in Julia [56].

Again, listings should not end sections. You should always describe what is going on in the displayed code after presenting a listing.

8.6 THEOREM BLOCKS

Example 8.1 and Esimerkki 8.2 show examples of theorem blocks:

Example 8.1 (Pythagorean theorem). For sides a, b and c of a right triangle, where c is the *hypotenuse*, the following equality holds: $a^2 + b^2 = c^2$.

Proof. Proof goes here. ∎

Esimerkki 8.2 (Pythagoraan lause). Suorakulmaisen kolmion sivut a, b ja c, missä c on *hypotenuusa*, toteuttavat yhtälön $a^2 + b^2 = c^2$.

Todistus. Todistus tulee tähän. ∎

As one can see, these follow the same counter. The theorem box has been typeset with the command seen in Listing 8.3.

```
#tauthesis.example(
    title: [Pythagorean theorem],
)[
    For sides
    #math.equation(
        alt: "a",
        $a$
    ),
    #math.equation(
        alt: "b",
        $b$
    )
    and
    #math.equation(
        alt: "c",
        $c$
    ),
    where
    #math.equation(
        alt: "c",
        $c$
    )
    is the _hypotenuse_, the following equality holds:
    #math.equation(
        alt: altTextFn("
            a^2 + b^2 = c^2 .
        "),
        $
            a^2 + b^2 = c^2 thin .
        $
    )
] <pythagorean-theorem>
```

Listing 8.3. Example code for displaying a theorem block.

The reference label is placed after the call to `tauthesis.example` via the `<pythagorean-theorem>` syntax. The module `tauthesis` defines the following theorem box functions:

- `definition`,
- `theorem`,
- `lemma`,
- `corollary` and
- `example`.

Their Finnish equivalents

- `määritelmä`,
- `lause`,
- `apulause`,

- `seurauslause` and
- `esimerkki`

are also defined. These all follow the same counter, and can all be referenced by suffixing the equivalent function call with a tag.

Try not to overuse these, however. Academic theses should *not* be written in the *Landau style*, where the entire text consists of definitions after definitions and theorems after theorems. There should always be a full paragraph between two theorems or other objects, such as figures and tables.

8.7 ACCESSIBILITY CONSIDERATIONS

In accordance with the European Union directive 2016/2102 [57], Finnish law [58] requires that all electronic publications of public sector bodies be made *accessible*, as long as doing so would not require an infeasible amount of work. This also applies to academic theses published by universities, including yours [59]. For this reason you should be aware of the requirements related to accessible electronic documents. Some of these considerations are described below. These are also discussed in the Typst accessibility guide [49].

Use of language: Use simple and descriptive language. Avoid overtly long sentences. Any truly essential information should be delivered in words and not by visual means.

Figures: As per the point above, any figures in the document should have some kind of a textual alternative available. Ideally so called *alternative texts* [60], [61], [62] (which are not the same thing as figure captions) should be included into a document. However, if this is not feasible due to current technological limitations, one should at least try to describe what is happening in each image or table in the body text as well as they can. The Typst functions `image` [48] and `figure` [50] accept the named argument `alt`, which allows you to provide an alternative text for an included image.

Document metadata: This should be as complete as possible. However, the most essential metadata fields are the *title* of the work, and the *main language* the work was written in. This is relevant from the point of view of assistive software like screen readers, that visually impaired consumers of a document might be relying on. Also, resources such as the fonts intended to be used with a document should be embedded into the document itself.

Mathematics: As professor Antti Valmari previously of Tampere University of Technology fame once said in a conversation with one of the maintainers of this template, "Mathematicians have an inexplicable urge to put things on top of each other.". From the point of view of accessibility of PDF files, this is a terrible practice, so long as the document format does not include either structural information related to the semantics of the visual presentation within it, or alternative texts that screen reader sofware can access. This is because classically PDF is a format where graphical elements are essentially drawn on a canvas with x- and y-coordinates, which becomes apparent if one tries to copy and paste the following fractions from the output PDF file of this template:

$$a \div b\,, \quad a/b\,, \quad \frac{a}{b} \quad \text{and} \quad ab^{-1}\,. \tag{8.6}$$

Depending on your PDF reader implementation, it might be that with the vertically stacked fraction $\frac{a}{b}$, the numerator a ends up on the previous line, since it is vertically above the line where the fraction itself resides, and reading order is based on the coordinates (elements higher and more to the left come first). Alternatively, it might be that the stacked fraction is handled as a whole, but since the horizontal fraction line separating the numerator a and denominator b is implemented as a graphical element, which does not have a textual representation in the PDF model, the line will be lost, and you get something like $a\,b$ as the output. This is semantically the complete opposite (or inverse) of the indended meaning of the fraction denominator!

The denominator $1 \div b$ expressed with a negative exponent b^{-1} is just as problematic, since upper and lower indices also don't generally have a proper textual representation in Unicode. When copying and pasting such an expression, you might end up with something like $ab - 1$ in the textual output. This also completely changes the semantics of the output expression.

As of October 2025, neither Typst or LaTeX currently support automated creation of so-called *well-tagged PDF files* [63], which are documents carrying structural information similar to HTML tags. Typst will start introducing support for automatically producing tagged PDF files starting from version 0.14.0 [64], [65]. LaTeX has some experimental support for producing tagged PDFs starting from TeX Live 2024 [66], but it requires intervention from the user of the language, such as manually inserting LaTeX commands to generate said tags. Another issue with well-tagged PDFs is that many PDF readers still do not support the standard. While we wait for the related technology such as the embedding of MathML [63], [67] into PDF files to catch up, there are some simple things you can do to make your thesis more accessible.

8.7.1 Add alternative texts or other structure to your equations

The importance of this was explained in Section 8.4.2. Typst version 0.14.0 supports adding alternative texts to equations through the function `math.equation` [55] named argument `alt`. In future versions on Typst, it is also intended to support the PDF/UA-2 standard, which would allow automatically embedding structural information in the form of MathML into a PDF document. This would remove the need for alternative texts in the case of equations.

Another means of keeping text accessible is to keep the flow of the text as horizontal as possible, which might allow text extraction programs such as `pdftotext` [68], which also operate on the coordinate system basis, to extract the text in a correct reading order. Paul Dirac did a similar thing with his bra–ket notation for linear algebra products. This enabled him to write mathematics with a normal chevron-equipped typewriter, without access to special symbols such as greek letters, upper and lower indices and integrals. Especially, it simplified writing long subindex labels related to linear operators such as integrals [69]:

$$\langle i \mid j \rangle \coloneqq \int \varphi_i * \psi_j \,\mathrm{d}x\,. \tag{8.7}$$

Here φ and ψ denote wave functions, and the left-associative unary operator $a\,*$ denotes the complex conjugate of its argument a. However, inventing your own notations should be done sparingly, since this is against the guideline of sticking to well-established notations. Inventing custom notations also carries with it the burden of having to explain yourself more.

8.7.2 Convert your final submission to PDF/UA-1 format

If you are using Typst version ≥ 0.14.0 locally via a command line, it is possible to produce a thesis into a tagged PDF/UA-1 format via the command

```
typst compile --pdf-standard ua-1 main.typ
```

If the file was generated correctly according to the PDF/UA-1 standard, it should contain accessibility tags. There are not many free tools that can actually visualize these tags for non-visually-impaired people, but the friendly people of the LaTeX Project have an online tool showtags [70] that you might use for this purpose. For the Microsoft Windows® platform, there also exists the tool PAC [71] or *PDF Accessibility Checker*. The PDF Association has also published a list of other tools that might be used to check PDF files for accessibility [72]. Before using either of these, run your thesis through the PDF validator veraPDF [73].

The standard also allows you to embed arbitrary files into your PDF file, using Typst's PDF module function `pdf.attach` [74], and adds accessibility tags to the document. As discussed below, Typst compilers before version 0.14.0 cannot yet produce truly accessible documents, so including the source code of your thesis chapters as plain text provides an alternative means of accessing your thesis. This template has tried to demonstrate how this can be performed at the beginning of each front- and main matter file, and in the example appendices. If you are using the Typst Web application, simply choose PDF/UA-1 as the output format in the file export menu.

Note: if you are unable to produce an accessible version of your thesis according to these PDF accessibility checkers, you should attach the source code of the thesis in a sensible reading order to the Trepo submission! This template shows examples of embedding the source code into the PDF file itself, but note that a separate ZIP archive is the best format for the source code submission. Visually impaired people might not be able to extract the attachments from a PDF file, but a separate ZIP archive can be opened easily on any system.

8.7.3 COMPILATION DISSERTATIONS AND PDF ATTACHMENTS

If you are writing an accessible compilation dissertation and wish to include scientific articles as attachments to your document[1], you might need to do it in post production. Typst can include PDF files via the `image` function, but most publishers at this point do not care about accessibility that much, and therefore the PDFs downloaded from their websites often do not contain accessibility tags.

The PDF renderer in the Typst compiler cannot handle this lack of accessibilty tags automatically, as there is practically no means of automatically deducing what the tag structure should have been in such a document. Therefore Typst does not support exporting documents in accessible PDF formats, if you attempt to import PDF images into your thesis. However, see Section 9.3 for how you might still do this, to at least preview what your final product might look like.

Note that if you published your work via *open access publishing channels* [75] with a permissive license [76], you can create a *derivative work* of your publication, such as an accessible Typst version of it, and attach that to your dissertation instead. However, also note that you should still refer to the peer-reviewed versions of your publications in the bibliography and on the attached publication title pages, such as is done in this template for the attached example publications (if the selected thesis type is "high" enough.)

[1] Including publications is not actually required for compilation PhD dissertations. Only referring to the publications via full bibliographic information is actually required.

9 References

Different referencing styles determine how you create in-text citations and the bibliography (list of references). Two common referencing styles are presented in this chapter:

- Numeric referencing (Vancouver system), such as [1], [2], …
- Name-year (Harvard) system, such as (Weber 2001), (Kaunisto 2003), …

A numeric reference is inserted in square brackets, whereas the last name of the author and the year of publication are given in parentheses. Both styles are acceptable, but the conventions for referencing vary between disciplines. You must pick one and use it consistently throughout your thesis. The list of possible styles can be found on the Typst documentation page [77].

The most commonly used tool for creating bibliographies in Typst is the Hayagriva [78] YAML [79] format, but Typst also supports BibLaTeX [80]. The former is what this thesis template uses by default. Both Hayagriva and BibLaTeX are based on gathering bibliographic information from the sources to a bibliography file using a specialised syntax. Typst reads both this file and the document being written, and then forms the references and the bibliography based on this information. A citation style can be chosen by setting it in the `metadata.typ` file, in the variable `citationStyle`.

9.1 In-text citations

In-text citations are placed within the body of the text as close to the actual claim that the citation supports as possible. The citation is generally placed within the sentence before the next period:

- Weber argues that… [1].
- Cattaneo et al. introduce in their study [2] a new…
- The result isldots [1, p. 23]. One must also note… [1, pp. 33–36]
- In accordance with the presented theory… (Weber 2001).
- It must especially be noted… (Cattaneo et al. 2004).
- Weber (2001, p. 230) has stated …
- Based on literature in the field [1, 3, 5] …
- Based on literature in the field [1][3][5] …
- The topic has been widely studied [6–18] …
- … existing literature (Weber 2001; Kaunisto 2003; Cattaneo et al. 2004) has …

You *can* also place a citation after a period, in which case it refers to the entire preceding paragraph. However, this is generally not recommended, as it might make it unclear which claim the citation supports, if there are manu within the preceding paragraph.

9.2 THE BIBLIOGRAPHY FILE

Typst works similarly to LaTeX when it comes to citations and the bibliography:

1. A user collects a database of references into a bibliography file, where each reference has a *unique* identifier string that can be referenced, such as `mainauthor-etal-topic-year`.
2. The file is imported into a project using the `bibliography` function [77] of Typst. This is done in the file `main.typ` in the case of this template.
3. To cite a bibliography file entry with the identifier `id`, one writes `@id` within the text of their Typst project. As an example of what this might look like, placing `@Bezanson2017Julia` in a paragraph results in the following citation: [56].

The bibliogprahy file formats that Typst supports are *Hayagriva* [78] and *BibLaTeX* [80]. This template contains mostly equivalent examples of both respective formats in the form of the files `bibliography.yaml` and `bibliography.bib`.

The preferred format for this template is Hayagriva, the reason for which will be explained in Section 9.3. If you have an existing bibliography file in the form of BibLaTeX, Hayagriva can also be installed as a command line program [78, Installation], and the BibLaTeX file converted to Hayagriva format using the command

```
hayagriva bibliography.bib > bibliography.yaml
```

There are different kinds of citation types such as articles or web pages, which should be typeset differently in the bibliography. In the case of Hayagriva, an article reference metadata is given in Listing 9.1, whereas an equivalent BibLaTeX entry is shown in Listing 9.2.

```yaml
Bezanson2017Julia:
  type: article
  title: >-
    Julia: A fresh approach to numerical computing
  author:
  - Bezanson, Jeff
  - Edelman, Alan
  - Karpinski, Stefan
  - Shah, Viral B
  date: 2017
  page-range: 65-98
  serial-number:
    doi: 10.1137/141000671
  url:
    value: https://doi.org/10.1137/141000671
    date: 2025-10-10
  parent:
    type: periodical
    title: SIAM review
    publisher: SIAM
    issue: 1
    volume: 59
```

Listing 9.1. An example of a Hayagriva article entry.

```bibtex
@article{
  Bezanson2017Julia,
  title={{Julia: A fresh approach to numerical computing}},
  author={Bezanson, Jeff and Edelman, Alan and Karpinski, Stefan and Shah, Viral B},
  journal={SIAM review},
  volume={59},
  number={1},
  pages={65--98},
  year={2017},
  publisher={SIAM},
  url={https://doi.org/10.1137/141000671}
}
```

Listing 9.2. An example of a BibLaTeX article entry.

It is often preferable to order the list of references alphabetically by the first author's last name. This template however uses the `ieee` style by default, meaning that citations are listed in the bibliography in the order of appearance. Change the value of the variable `citationStyle` in the file `metadata.typ` to change this.

9.3 INSERTING PHD PUBLICATIONS INTO YOUR THESIS

As was mentioned in Section 8.7.3, the PDF format has some limitations regarding attaching PDF files into other PDF files, if one wishes for the final product to remain standard-conforming. Regardless, this template provides support for inserting PhD publications at the end of this document[2] using the bibliography file `bibliography.yaml`, if the export format is not one of the tagged variants such as PDF/A-3a or PDF/UA-1. You can therefore test how the PDF attachments would look like if you just compile your document without accessibility tags:

```
typst compile template/main.typ
```

To tell the template which of your references in `bibliography.yaml` should be considered as compilation dissertation attachments, you should add additional fields to the Hayagriva file `bibliography.yaml` entries, that correspond to your PhD publications. These are of the following form:

```yaml
path: publications/publication.pdf
license: null or the name of the license
n-of-pages: the number of pages in the document
tauthesis-publication: true or false
author-contribution: >
  A short description of how you contributed to this work.
  This can extend on multiple lines, as long as the lines
  are indented.
```

The `path` field should be in relation to the `main.typ` file. It might be a good idea to utilize the readily provided `publications/` folder in the template directory, where the example publications already reside.

9.4 ATTACHING TYPST-BASED PHD PUBLICATIONS

If you are in the lucky position of being allowed to use Typst source code to enter your dissertation publications into your thesis[3], it should be entirely possible to make your entire dissertation accessible. To aid with this, targeting *Open Access publishing channels* [75] and preferring licenses such as CC BY 4.0 [81] in your publications allows you to create a *derivative work* from the publication. A derivative work can be a Typst version of the publication.

The Typst source simply needs to adhere to related accessibility standard requirements [49], [59]. Simply point the `path` field of a Hayagriva bibliography entry to a valid Typst root file of your publication, and this template will `#include` it.

[2]Which is not compulsory for it to be accepted.
[3]Which you again do not *need* to do. It is just a bonus if you can.

Note 1: Any included Typst files should *not* set their own page size or margins in a way which overwrites the ones of this template, as the page sizes need to match. Remove any

```
#set page(...)
```

invocations from the start of the publication file, and let this template handle the page settings for you. Otherwise you are allowed to customize the appearance of the publications using `set` and `show` rules of Typst, unless you wish to use the style of this template. Any utilized fonts should be accessible, though, so keep that in mind.

Note 2: Typst does not support multiple bibliographies in version 0.14.0 natively, but it is a planned feature of version 0.15.0. In the meanwhile, you might wish to utilize a package such as `alexandria` or `pergamon` to typeset the bibliography of an attached Typst-based PhD publication and keep it separate from the bibliography of the main dissertation.

10 Conclusions

This template and the writing guidelines should help achieving consistently formatted and clear documents. Every writing and presentation must have a conclusion. This fact is here emphasized by having this short and rather artificial summary also in this template. The purpose of this chapter is to discuss how the things we set out to do in the introduction succeeded, and compare our results to other literature on the same topic.

As a last reminder, remember to make sure that your document is *accessible* [49] before submitting it to the Tampere University archive Trepo. See the university library guidelines [82] regarding this. Use veraPDF [73], showtags [70], PAC [71] (if you have access to Windows®) or some other software listed by the PDF Association [72] to verify the thesis PDF document accessibility, before your thesis submission.

If your thesis PDF file is not accessible and you have no way of making it so, you should attach both the unaccessible PDF file at least in PDF/A-2b format, and your *thesis source code* as a ZIP file to the Trepo submission. Be sure to clean up any personal comments from the source code before doing so. Also make sure that the final submitted source code works, meaning a PDF file can be compiled from it.

Note that you are allowed to attach the source code and a permissive license such as CC BY 4.0 [81] to your thesis, even if it is accessible. This is *the* ideal way of publishing your thesis, unless you are restricted from doing so by a company or other entity tied to your thesis process, that migth require secrecy from you. The authors of this template and the Tampere University Library recommend CC-licensing as the ideal licensing option [76], in the name of promoting open science.

11 Test

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Appendix A: Subtype-specific GO Enrichment Biological Processes

A.1 Classical

Keywords used:

```
classical_keywords = [
    "cell cycle",
    "cell division",
    "mitotic",
    "DNA replication",
    "DNA repair",
    "chromosome",
    "proliferation",
    "checkpoint",
    "EGFR",
    "growth factor",
    "MAPK",
    "ERK",
    "cell migration",
    "cell motility"
]
```

Output:

Rank	Term	Adjusted P-value
3	Regulation Of Cell Migration (GO:0030334)	3.995484e-13
22	Positive Regulation Of Cell Migration (GO:0030335)	1.108954e-07
36	Regulation Of ERK1 And ERK2 Cascade (GO:0070372)	3.583929e-06
52	Positive Regulation Of Cell Motility (GO:2000147)	1.361010e-05
63	Negative Regulation Of Cell Migration (GO:0030336)	2.530444e-05
68	Positive Regulation Of Cell Population Proliferation (GO:0008284)	3.776415e-05
75	Cellular Response To Growth Factor Stimulus (GO:0071363)	4.698021e-05
77	Negative Regulation Of Cell Motility (GO:2000146)	5.145454e-05
85	Positive Regulation Of Epithelial Cell Migration (GO:0010634)	7.747226e-05
111	Regulation Of Endothelial Cell Migration (GO:0010594)	1.786491e-04
114	Positive Regulation Of Endothelial Cell Migration (GO:0010595)	2.037032e-04
131	DNA-templated DNA Replication (GO:0006261)	2.792774e-04
161	Positive Regulation Of Epithelial Cell Proliferation (GO:0050679)	5.232011e-04
168	Regulation Of Cell Population Proliferation (GO:0042127)	5.502765e-04
173	Positive Regulation Of ERK1 And ERK2 Cascade (GO:0070374)	5.849589e-04
180	Negative Regulation Of Smooth Muscle Cell Migration (GO:0014912)	7.361593e-04
204	Positive Regulation Of MAPK Cascade (GO:0043410)	1.070676e-03
207	Mitotic DNA Replication (GO:1902969)	1.181616e-03
219	G2/M Transition Of Mitotic Cell Cycle (GO:0000086)	1.358285e-03

Rank	Term	Adjusted P-value
237	Positive Regulation Of Chromosome Segregation (GO:0051984)	1.882370e-03
268	Cell Cycle G2/M Phase Transition (GO:0044839)	2.549435e-03
274	Regulation Of Smooth Muscle Cell Proliferation (GO:0048660)	2.871201e-03
281	Regulation Of Smooth Muscle Cell Migration (GO:0014910)	3.141319e-03
291	Regulation Of Mononuclear Cell Migration (GO:0071675)	3.306056e-03
294	Regulation Of Cell Motility (GO:2000145)	3.391214e-03
298	Cellular Response To Transforming Growth Factor Beta Stimulus (GO:0071560)	3.717364e-03
318	Positive Regulation Of Neural Precursor Cell Proliferation (GO:2000179)	4.826338e-03
323	DNA Unwinding Involved In DNA Replication (GO:0006268)	5.024437e-03
332	DNA Strand Elongation Involved In DNA Replication (GO:0006271)	5.639352e-03
339	Negative Regulation Of Cell Population Proliferation (GO:0008285)	6.208020e-03
344	Mitotic Sister Chromatid Segregation (GO:0000070)	6.240460e-03
348	Regulation Of Blood Vessel Endothelial Cell Migration (GO:0043535)	6.746236e-03
352	Endothelial Cell Migration (GO:0043542)	6.765134e-03
358	Positive Regulation Of Smooth Muscle Cell Proliferation (GO:0048661)	6.817300e-03
362	DNA Replication (GO:0006260)	7.343744e-03
365	Positive Regulation Of Neuroblast Proliferation (GO:0002052)	7.497843e-03
368	Negative Regulation Of Smooth Muscle Cell Proliferation (GO:0048662)	7.497843e-03
371	Positive Regulation Of Endothelial Cell Proliferation (GO:0001938)	7.539944e-03
385	Regulation Of Mitotic Cell Cycle Phase Transition (GO:1901990)	9.620599e-03
403	Mitotic Cell Cycle Phase Transition (GO:0044772)	1.129297e-02
410	Regulation Of Fibroblast Growth Factor Receptor Signaling Pathway (GO:0040036)	1.179074e-02
420	Positive Regulation Of Blood Vessel Endothelial Cell Migration (GO:0043536)	1.276006e-02
424	Negative Regulation Of Lymphocyte Proliferation (GO:0050672)	1.303175e-02
437	Neural Crest Cell Migration (GO:0001755)	1.483222e-02
442	Negative Regulation Of MAPK Cascade (GO:0043409)	1.510820e-02
446	Regulation Of Endothelial Cell Proliferation (GO:0001936)	1.533544e-02
456	Establishment Of Chromosome Localization (GO:0051303)	1.736457e-02
465	MAPK Cascade (GO:0000165)	1.761472e-02
469	Positive Regulation Of Alpha-Beta T Cell Proliferation (GO:0046641)	1.763428e-02
481	Negative Regulation Of ERK1 And ERK2 Cascade (GO:0070373)	1.972286e-02
505	Regulation Of Neuroblast Proliferation (GO:1902692)	2.281059e-02
514	Response To Transforming Growth Factor Beta (GO:0071559)	2.412420e-02
524	DNA Replication Initiation (GO:0006270)	2.422485e-02
534	Regulation Of Cell Cycle Process (GO:0010564)	2.478471e-02
539	Regulation Of Chromosome Separation (GO:1905818)	2.525723e-02
541	Stress-Activated MAPK Cascade (GO:0051403)	2.527638e-02
547	Regulation Of Cell Cycle G2/M Phase Transition (GO:1902749)	2.527638e-02
560	Positive Regulation Of Fibroblast Proliferation (GO:0048146)	2.619130e-02
562	Mitotic Spindle Organization (GO:0007052)	2.619130e-02
569	Regulation Of Epidermal Growth Factor Receptor Signaling Pathway (GO:0042058)	2.645629e-02
570	Regulation Of G1/S Transition Of Mitotic Cell Cycle (GO:2000045)	2.805140e-02
576	Mesenchymal Cell Migration (GO:0090497)	2.994402e-02
577	Regulation Of Metaphase/Anaphase Transition Of Cell Cycle (GO:1902099)	2.994402e-02
583	Positive Regulation Of Cell Migration By Vascular Endothelial Growth Factor Signaling Pathway (GO:0038089)	2.994402e-02
584	Positive Regulation Of Chromosome Condensation (GO:1905821)	2.994402e-02
590	Regulation Of Chromosome Condensation (GO:0060623)	2.994402e-02
601	Blood Vessel Endothelial Cell Migration (GO:0043534)	3.228637e-02

Rank	Term	Adjusted P-value
612	Mitotic Cytokinesis (GO:0000281)	3.547076e-02
632	Nuclear Chromosome Segregation (GO:0098813)	3.754081e-02
635	Positive Regulation Of Smooth Muscle Cell Migration (GO:0014911)	3.754081e-02
641	Mitotic Spindle Assembly Checkpoint Signaling (GO:0007094)	3.810509e-02
642	Mitotic Spindle Checkpoint Signaling (GO:0071174)	3.810509e-02
643	Spindle Assembly Checkpoint Signaling (GO:0071173)	3.810509e-02
648	Positive Regulation Of Chromosome Separation (GO:1905820)	3.810509e-02
668	Positive Regulation Of Osteoblast Proliferation (GO:0033690)	3.972481e-02
678	Transforming Growth Factor Beta Receptor Signaling Pathway (GO:0007179)	3.973461e-02
689	Mitotic Nuclear Division (GO:0140014)	4.421790e-02

A.2 MESENCHYMAL

Keywords used:

```
mesenchymal_keywords = [
    "immune",
    "inflamm",
    "cytok",
    "interleukin",
    "interferon",
    "leukocyte",
    "lymphocyte",
    "macrophage",
    "chemokine",
    "chemotaxis",
    "extracellular",
    "matrix",
    "collagen",
    "cell adhesion",
    "wound",
    "angiogenesis",
    "vasculature",
    "fibroblast",
    "epithelial",
    "mesenchymal"
]
```

Output:

Rank	Term	Adjusted P-value
11	Regulation Of Cell Adhesion (GO:0030155)	2.191577e-09
16	Positive Regulation Of Cytokine Production (GO:0001819)	1.765843e-08
19	Response To Cytokine (GO:0034097)	4.534666e-08
24	Cell-Matrix Adhesion (GO:0007160)	1.016677e-07
25	Positive Regulation Of Tumor Necrosis Factor Superfamily Cytokine Production (GO:1903557)	1.016677e-07
42	Collagen Fibril Organization (GO:0030199)	6.110994e-07

Rank	Term	Adjusted P-value
44	Positive Regulation Of Inflammatory Response (GO:0050729)	1.086884e-06
49	Inflammatory Response (GO:0006954)	1.586640e-06
54	Regulation Of Interleukin-6 Production (GO:0032675)	3.193197e-06
55	Wound Healing, Spreading Of Cells (GO:0044319)	3.475930e-06
76	Extracellular Matrix Organization (GO:0030198)	1.677365e-05
79	Cellular Response To Cytokine Stimulus (GO:0071345)	2.179767e-05
90	Regulation Of Inflammatory Response (GO:0050727)	4.293276e-05
100	Cell-Cell Adhesion Via Plasma-Membrane Adhesion Molecules (GO:0098742)	6.317949e-05
103	Positive Regulation Of Interleukin-6 Production (GO:0032755)	6.676476e-05
107	Response To Type II Interferon (GO:0034341)	8.561360e-05
108	Macrophage Activation (GO:0042116)	8.781421e-05
117	Regulation Of Interleukin-1 Beta Production (GO:0032651)	1.010323e-04
119	Regulation Of Interleukin-8 Production (GO:0032677)	1.091639e-04
120	Regulation Of Angiogenesis (GO:0045765)	1.099990e-04
136	Leukocyte Cell-Cell Adhesion (GO:0007159)	1.679717e-04
152	Positive Regulation Of Interleukin-8 Production (GO:0032757)	2.064958e-04
155	Heterotypic Cell-Cell Adhesion (GO:0034113)	2.281749e-04
161	Positive Regulation Of Cell Adhesion (GO:0045785)	2.660581e-04
162	Regulation Of Cell-Matrix Adhesion (GO:0001952)	2.660581e-04
171	Negative Regulation Of Leukocyte Activation (GO:0002695)	3.229880e-04
173	Positive Regulation Of Cytokine-Mediated Signaling Pathway (GO:0001961)	3.253548e-04
176	Cellular Response To Type II Interferon (GO:0071346)	3.469532e-04
183	Regulation Of Macrophage Derived Foam Cell Differentiation (GO:0010743)	3.636784e-04
191	Negative Regulation Of Lymphocyte Proliferation (GO:0050672)	4.550607e-04
217	Homotypic Cell-Cell Adhesion (GO:0034109)	6.356537e-04
221	Regulation Of Cytokine-Mediated Signaling Pathway (GO:0001959)	6.364527e-04
224	Negative Regulation Of Cell Adhesion (GO:0007162)	6.938003e-04
226	Positive Regulation Of Chemokine Production (GO:0032722)	8.308837e-04
227	Positive Regulation Of Lymphocyte Migration (GO:2000403)	8.571335e-04
229	Positive Regulation Of Epithelial To Mesenchymal Transition (GO:0010718)	9.156479e-04
230	Positive Regulation Of Leukocyte Cell-Cell Adhesion (GO:1903039)	9.387090e-04
234	Cytokine-Mediated Signaling Pathway (GO:0019221)	1.064162e-03
239	Positive Regulation Of Interleukin-1 Production (GO:0032732)	1.225762e-03
245	Regulation Of Cell Adhesion Mediated By Integrin (GO:0033628)	1.403743e-03
259	Positive Regulation Of Macrophage Derived Foam Cell Differentiation (GO:0010744)	1.735484e-03
262	Negative Regulation Of Leukocyte Degranulation (GO:0043301)	1.823332e-03
266	Regulation Of Epithelial To Mesenchymal Transition (GO:0010717)	1.977209e-03
267	Positive Regulation Of Angiogenesis (GO:0045766)	2.005749e-03
281	Epiboly Involved In Wound Healing (GO:0090505)	2.358125e-03
284	Regulation Of Immune Response (GO:0050776)	2.410616e-03
287	Positive Regulation Of Epithelial Cell Migration (GO:0010634)	2.596269e-03
289	Positive Regulation Of Interleukin-1 Beta Production (GO:0032731)	2.667769e-03
291	Regulation Of Chemokine Production (GO:0032642)	2.796806e-03
313	Regulation Of Macrophage Cytokine Production (GO:0010935)	3.076783e-03
331	Negative Regulation Of Immune Response (GO:0050777)	4.581672e-03
338	Positive Regulation Of Cell-Matrix Adhesion (GO:0001954)	5.043811e-03
341	Negative Regulation Of Myeloid Leukocyte Mediated Immunity (GO:0002887)	5.043811e-03
351	Regulation Of Inflammatory Response To Wounding (GO:0106014)	5.043811e-03

Rank	Term	Adjusted P-value
352	Wound Healing, Spreading Of Epidermal Cells (GO:0035313)	5.043811e-03
354	Interleukin-1-Mediated Signaling Pathway (GO:0070498)	5.161696e-03
356	Positive Regulation Of Myeloid Leukocyte Differentiation (GO:0002763)	5.350819e-03
357	Cellular Response To Interleukin-1 (GO:0071347)	5.422992e-03
358	Homophilic Cell Adhesion Via Plasma Membrane Adhesion Molecules (GO:0007156)	5.478076e-03
364	Negative Regulation Of Inflammatory Response To Wounding (GO:0106015)	5.602314e-03
365	Negative Regulation Of Leukocyte Proliferation (GO:0070664)	5.602314e-03
366	Positive Regulation Of Cardiac Epithelial To Mesenchymal Transition (GO:0062043)	5.602314e-03
367	Positive Regulation Of Fibroblast Migration (GO:0010763)	5.602314e-03
370	Regulation Of Epithelial To Mesenchymal Transition Involved In Endocardial Cushion Formation (GO:1905005)	5.602314e-03
376	Negative Regulation Of Response To Wounding (GO:1903035)	5.858995e-03
385	Positive Regulation Of Type I Interferon Production (GO:0032481)	5.959048e-03
386	Positive Regulation Of Vasculature Development (GO:1904018)	6.370565e-03
389	Positive Regulation Of Chemotaxis (GO:0050921)	6.456837e-03
400	Regulation Of Fibroblast Migration (GO:0010762)	8.209142e-03
401	Positive Regulation Of Neuroinflammatory Response (GO:0150078)	8.209142e-03
408	Regulation Of Extracellular Matrix Organization (GO:1903053)	8.689382e-03
427	Epithelial Cell Migration (GO:0010631)	1.133202e-02
433	Positive Regulation Of Epithelial Cell Proliferation (GO:0050679)	1.184297e-02
434	Regulation Of Innate Immune Response (GO:0045088)	1.211593e-02
439	Regulation Of Vasculature Development (GO:1901342)	1.235661e-02
440	Myeloid Leukocyte Differentiation (GO:0002573)	1.270481e-02
448	Regulation Of Cytokine Production (GO:0001817)	1.301922e-02
449	Regulation Of Interferon-Beta Production (GO:0032648)	1.302270e-02
464	Immune Response-Regulating Cell Surface Receptor Signaling Pathway (GO:0002768)	1.520886e-02
468	Negative Regulation Of Immune System Process (GO:0002683)	1.602853e-02
471	Negative Regulation Of Cell-Matrix Adhesion (GO:0001953)	1.705836e-02
481	Positive Regulation Of Homotypic Cell-Cell Adhesion (GO:0034112)	1.732944e-02
482	Positive Regulation Of Myeloid Leukocyte Cytokine Production Involved In Immune Response (GO:0061081)	1.732944e-02
495	Glomerular Epithelial Cell Development (GO:0072310)	1.761170e-02
505	Leukocyte Adhesion To Vascular Endothelial Cell (GO:0061756)	1.804215e-02
532	Epithelial Cell-Cell Adhesion (GO:0090136)	2.044802e-02
533	Macrophage Activation Involved In Immune Response (GO:0002281)	2.044802e-02
538	Neutrophil Activation Involved In Immune Response (GO:0002283)	2.044802e-02
566	Negative Regulation Of Leukocyte Mediated Cytotoxicity (GO:0001911)	2.380586e-02
569	Positive Regulation Of Endothelial Cell Chemotaxis (GO:2001028)	2.380586e-02
570	Regulation Of Endothelial Cell Chemotaxis (GO:2001026)	2.380586e-02
581	Positive Regulation Of Interferon-Beta Production (GO:0032728)	2.516337e-02
583	Leukocyte Tethering Or Rolling (GO:0050901)	2.516337e-02
584	Myeloid Cell Activation Involved In Immune Response (GO:0002275)	2.516337e-02
586	Positive Regulation Of Macrophage Activation (GO:0043032)	2.516337e-02
591	Regulation Of Interleukin-12 Production (GO:0032655)	2.617121e-02
597	Positive Regulation Of Interleukin-12 Production (GO:0032735)	2.760463e-02
598	Regulation Of Type I Interferon-Mediated Signaling Pathway (GO:0060338)	2.760463e-02
603	Positive Regulation Of Leukocyte Apoptotic Process (GO:2000108)	2.797329e-02
611	Establishment Or Maintenance Of Epithelial Cell Apical/Basal Polarity (GO:0045197)	2.892257e-02
614	Wound Healing (GO:0042060)	2.892257e-02
615	Regulation Of Interleukin-2 Production (GO:0032663)	2.900581e-02

Rank	Term	Adjusted P-value
617	Negative Regulation Of Inflammatory Response (GO:0050728)	2.944249e-02
618	Immunoglobulin Mediated Immune Response (GO:0016064)	2.962474e-02
619	Innate Immune Response Activating Cell Surface Receptor Signaling Pathway (GO:0002220)	2.962474e-02
624	Granulocyte Chemotaxis (GO:0071621)	3.171126e-02
627	Positive Regulation Of Immune Effector Process (GO:0002699)	3.211490e-02
643	Positive Regulation Of Myeloid Leukocyte Mediated Immunity (GO:0002888)	3.383088e-02
646	Regulation Of Leukocyte Adhesion To Vascular Endothelial Cell (GO:1904994)	3.383088e-02
652	Response To Interleukin-1 (GO:0070555)	3.419055e-02
657	Neutrophil Chemotaxis (GO:0030593)	3.516418e-02
668	Mitotic Cytokinesis (GO:0000281)	3.549577e-02
680	Positive Regulation Of Macrophage Migration (GO:1905523)	3.555303e-02
685	Extracellular Structure Organization (GO:0043062)	3.623234e-02
687	Negative Regulation Of Cytokine Production (GO:0001818)	3.643416e-02
689	Leukocyte Aggregation (GO:0070486)	3.643416e-02
699	Regulation Of Leukocyte Degranulation (GO:0043300)	3.643416e-02
700	Regulation Of Lymphocyte Apoptotic Process (GO:0070228)	3.643416e-02
701	Regulation Of Positive Chemotaxis (GO:0050926)	3.643416e-02
705	Negative Chemotaxis (GO:0050919)	3.651408e-02
716	Negative Regulation Of Wound Healing (GO:0061045)	4.041233e-02
727	Regulation Of Neutrophil Chemotaxis (GO:0090022)	4.419582e-02
760	Humoral Immune Response Mediated By Circulating Immunoglobulin (GO:0002455)	4.671813e-02
761	Interleukin-27-Mediated Signaling Pathway (GO:0070106)	4.671813e-02
779	Immunoglobulin Production Involved In Immunoglobulin-Mediated Immune Response (GO:0002381)	4.800033e-02
783	Regulation Of Fibroblast Growth Factor Receptor Signaling Pathway (GO:0040036)	4.800033e-02
784	Regulation Of Granulocyte Chemotaxis (GO:0071622)	4.800033e-02
786	Regulation Of Neuroinflammatory Response (GO:0150077)	4.800033e-02
793	Positive Regulation Of Leukocyte Degranulation (GO:0043302)	4.887885e-02
796	Regulation Of Leukocyte Migration (GO:0002685)	4.887885e-02
800	Negative Regulation Of Cytokine Production Involved In Immune Response (GO:0002719)	4.887885e-02
801	Negative Regulation Of Inflammatory Response To Antigenic Stimulus (GO:0002862)	4.887885e-02
803	Positive Regulation Of Positive Chemotaxis (GO:0050927)	4.887885e-02

A.3 PRONEURAL

Keywords used:

```
proneural_keywords = [
    "neuro",
    "neuron",
    "synap",
    "axon",
    "dendrite",
    "gliogenesis",
    "glial",
    "oligodendro",
    "myelin",
    "nervous system",
    "brain",
```

```
    "projection",
    "neurotransmitter",
    "differentiation",
    "development",
    "stem cell",
    "cell fate",
    "forebrain",
    "cerebral",
    "central nervous system"
]
```

Output:

Rank	Term	Adjusted P-value
1	Axonogenesis (GO:0007409)	8.463008e-13
2	Modulation Of Chemical Synaptic Transmission (GO:0050804)	1.186011e-12
3	Chemical Synaptic Transmission (GO:0007268)	3.091247e-12
4	Synapse Organization (GO:0050808)	3.619502e-10
5	Regulation Of Neuron Projection Development (GO:0010975)	3.619502e-10
6	Nervous System Development (GO:0007399)	1.096961e-09
7	Neurotransmitter Secretion (GO:0007269)	1.916698e-09
10	Neuron Projection Morphogenesis (GO:0048812)	2.940480e-09
11	Positive Regulation Of Synaptic Transmission (GO:0050806)	6.395368e-09
12	Signal Release From Synapse (GO:0099643)	1.329384e-08
14	Regulation Of Neurotransmitter Receptor Activity (GO:0099601)	3.477924e-08
16	Neuron Projection Guidance (GO:0097485)	6.537164e-08
18	Regulation Of Synaptic Transmission, Glutamatergic (GO:0051966)	8.219629e-08
21	Neuron Projection Development (GO:0031175)	1.433888e-07
24	Positive Regulation Of Cell Projection Organization (GO:0031346)	2.335599e-07
25	Axon Guidance (GO:0007411)	3.319791e-07
26	Central Nervous System Development (GO:0007417)	3.542255e-07
27	Neuron Development (GO:0048666)	4.082145e-07
30	Positive Regulation Of Neuron Projection Development (GO:0010976)	4.447000e-07
34	Regulation Of Neuronal Synaptic Plasticity (GO:0048168)	9.820382e-07
35	Anterograde Trans-Synaptic Signaling (GO:0098916)	1.289307e-06
40	Synaptic Vesicle Exocytosis (GO:0016079)	2.148965e-06
45	Axon Development (GO:0061564)	3.689468e-06
54	Cell Morphogenesis Involved In Neuron Differentiation (GO:0048667)	9.019468e-06
58	Positive Regulation Of Excitatory Postsynaptic Potential (GO:2000463)	1.287225e-05
67	Modulation Of Excitatory Postsynaptic Potential (GO:0098815)	2.295859e-05
69	Synaptic Vesicle Endocytosis (GO:0048488)	2.383906e-05
71	Synapse Assembly (GO:0007416)	3.752276e-05
75	Brain Development (GO:0007420)	4.293145e-05
77	Regulation Of Synapse Assembly (GO:0051963)	4.368148e-05
84	Regulation Of Dendritic Cell Differentiation (GO:2001198)	6.074960e-05
85	Neurotransmitter Transport (GO:0006836)	6.161246e-05
94	Regulation Of Postsynapse Organization (GO:0099175)	9.067362e-05
101	Microglial Cell Activation (GO:0001774)	1.173739e-04
102	Synaptic Vesicle Recycling (GO:0036465)	1.225951e-04
109	Positive Regulation Of Synapse Assembly (GO:0051965)	1.601976e-04
115	Vesicle-Mediated Transport In Synapse (GO:0099003)	1.731215e-04

Rank	Term	Adjusted P-value
116	Plasma Membrane Bounded Cell Projection Organization (GO:0120036)	1.734956e-04
133	Positive Regulation Of Developmental Growth (GO:0048639)	2.901806e-04
137	Synaptic Vesicle Cycle (GO:0099504)	3.385978e-04
141	Synaptic Transmission, GABAergic (GO:0051932)	3.716225e-04
144	Positive Regulation Of Synaptic Transmission, Glutamatergic (GO:0051968)	3.716225e-04
147	Regulation Of Neuron Death (GO:1901214)	3.716225e-04
149	Presynaptic Endocytosis (GO:0140238)	3.716225e-04
150	Regulation Of Long-Term Synaptic Potentiation (GO:1900271)	3.716225e-04
157	Positive Regulation Of Dendrite Extension (GO:1903861)	5.001363e-04
165	Endothelial Cell Development (GO:0001885)	5.761456e-04
167	Long-Term Synaptic Potentiation (GO:0060291)	5.761456e-04
168	Regulation Of Neuron Migration (GO:2001222)	6.046363e-04
179	Synapse Pruning (GO:0098883)	7.880849e-04
182	Regulation Of Dendrite Extension (GO:1903859)	9.281525e-04
183	Response To Axon Injury (GO:0048678)	9.281525e-04
187	Positive Regulation Of Cell Differentiation (GO:0045597)	1.076656e-03
200	Negative Regulation Of Cell Differentiation (GO:0045596)	1.374952e-03
202	Endodermal Cell Differentiation (GO:0035987)	1.382432e-03
209	Positive Regulation Of Neurogenesis (GO:0050769)	1.502638e-03
235	Plasma Membrane Bounded Cell Projection Morphogenesis (GO:0120039)	2.190213e-03
237	Calcium Ion-Regulated Exocytosis Of Neurotransmitter (GO:0048791)	2.198101e-03
269	Anterograde Axonal Protein Transport (GO:0099641)	2.649976e-03
273	Protein Localization To Presynapse (GO:1905383)	2.649976e-03
276	Regulation Of Macrophage Derived Foam Cell Differentiation (GO:0010743)	2.919435e-03
303	Positive Regulation Of Axonogenesis (GO:0050772)	4.063491e-03
304	Regulation Of Neurotransmitter Secretion (GO:0046928)	4.063491e-03
309	Regulation Of Axonogenesis (GO:0050770)	4.576400e-03
313	Regulation Of Plasma Membrane Bounded Cell Projection Organization (GO:0120035)	4.676086e-03
316	Dendrite Morphogenesis (GO:0048813)	4.790770e-03
320	Regulation Of Long-Term Neuronal Synaptic Plasticity (GO:0048169)	4.793170e-03
322	Regulation Of Synapse Organization (GO:0050807)	5.302495e-03
329	Inhibitory Synapse Assembly (GO:1904862)	6.067183e-03
331	Regulation Of Short-Term Neuronal Synaptic Plasticity (GO:0048172)	6.067183e-03
332	Positive Regulation Of Vasculature Development (GO:1904018)	6.214021e-03
340	Regulation Of Trans-Synaptic Signaling (GO:0099177)	6.471681e-03
346	Myeloid Leukocyte Differentiation (GO:0002573)	6.723355e-03
381	Cell Projection Organization (GO:0030030)	9.167989e-03
397	Neurotransmitter Receptor Transport, Postsynaptic Endosome To Lysosome (GO:0098943)	9.207130e-03
398	Regulation Of Microglial Cell Migration (GO:1904139)	9.207130e-03
401	Positive Regulation Of Nervous System Development (GO:0051962)	9.492635e-03
405	Positive Regulation Of Macrophage Derived Foam Cell Differentiation (GO:0010744)	9.581355e-03
407	Positive Regulation Of Neuron Death (GO:1901216)	9.581355e-03
420	Central Nervous System Neuron Differentiation (GO:0021953)	1.140119e-02
425	Regulation Of Dendrite Morphogenesis (GO:0048814)	1.140119e-02
427	T Cell Differentiation (GO:0030217)	1.140119e-02
430	Mononuclear Cell Differentiation (GO:1903131)	1.140119e-02
439	Anterograde Axonal Transport (GO:0008089)	1.352378e-02
447	Excitatory Synapse Assembly (GO:1904861)	1.404620e-02
451	Regulation Of Granulocyte Differentiation (GO:0030852)	1.404620e-02
453	Regulation Of Postsynaptic Neurotransmitter Receptor Activity (GO:0098962)	1.404620e-02

Rank	Term	Adjusted P-value
466	Regulation Of Nervous System Process (GO:0031644)	1.422257e-02
473	Axonal Fasciculation (GO:0007413)	1.541468e-02
474	Cell Projection Assembly (GO:0030031)	1.541468e-02
479	Regulation Of Osteoblast Differentiation (GO:0045667)	1.573248e-02
480	Skin Development (GO:0043588)	1.579054e-02
482	Positive Regulation Of Myeloid Leukocyte Differentiation (GO:0002763)	1.670331e-02
485	Generation Of Neurons (GO:0048699)	1.736230e-02
495	Ventricular Septum Development (GO:0003281)	2.086318e-02
501	Regulation Of B Cell Differentiation (GO:0045577)	2.112966e-02
502	Regulation Of Megakaryocyte Differentiation (GO:0045652)	2.112966e-02
515	Neurotransmitter Receptor Diffusion Trapping (GO:0099628)	2.112966e-02
518	Postsynaptic Neurotransmitter Receptor Diffusion Trapping (GO:0098970)	2.112966e-02
536	Axon Extension (GO:0048675)	2.460512e-02
550	Neuron Projection Fasciculation (GO:0106030)	2.460512e-02
559	Negative Regulation Of Cell Projection Organization (GO:0031345)	2.475368e-02
575	Myeloid Cell Development (GO:0061515)	2.695387e-02
589	Regulation Of Axon Extension (GO:0030516)	2.970775e-02
594	Negative Regulation Of Neuron Migration (GO:2001223)	2.970775e-02
595	Positive Regulation Of CD4-positive, CD25-positive, Alpha-Beta Regulatory T Cell Differentiation (GO:0032831)	2.970775e-02
597	Positive Regulation Of Granulocyte Differentiation (GO:0030854)	2.970775e-02
598	Positive Regulation Of Microglial Cell Migration (GO:1904141)	2.970775e-02
609	Neuron Projection Extension (GO:1990138)	2.990366e-02
616	Central Nervous System Neuron Development (GO:0021954)	3.210567e-02
622	Circulatory System Development (GO:0072359)	3.274801e-02
633	Regulation Of Neuron Apoptotic Process (GO:0043523)	3.579084e-02
638	Heart Development (GO:0007507)	3.579084e-02
648	Synaptic Vesicle Membrane Organization (GO:0048499)	3.579084e-02
650	Positive Regulation Of Osteoblast Differentiation (GO:0045669)	3.592668e-02
662	Negative Regulation Of Biomineral Tissue Development (GO:0070168)	3.690473e-02
663	Negative Regulation Of Osteoclast Differentiation (GO:0045671)	3.690473e-02
666	Regulation Of Neurotransmitter Transport (GO:0051588)	3.690473e-02
670	Negative Regulation Of Long-Term Synaptic Potentiation (GO:1900272)	3.720601e-02
674	Positive Regulation Of Megakaryocyte Differentiation (GO:0045654)	3.720601e-02
694	Neuron Differentiation (GO:0030182)	4.150973e-02
702	Regulation Of Dendrite Development (GO:0050773)	4.297896e-02
706	Positive Regulation Of Neuroblast Proliferation (GO:0002052)	4.297896e-02
719	Negative Regulation Of Neuron Projection Development (GO:0010977)	4.689814e-02
728	Positive Regulation Of Fat Cell Differentiation (GO:0045600)	4.878617e-02

Appendix B: Tests related to cross-referencing

This attachment is here just to make sure that cross-references work as intended. Towards this end, Figure B.1 is here just to make testing within the same chapter possible.

Figure B.1. A test image with a very long caption: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo.

Also, here is an equation for similar purposes:

$$\mathbf{L}\mathbf{x} \approx \mathbf{y}\,. \tag{B.1}$$

Also, why not add a theorem to the mix.

Theorem B.1 (Pythagorean theorem). For sides a, b and c of a right triangle, where c is the *hypotenuse*, the following equality holds: $a^2 + b^2 = c^2$.

B.1 Figure tests

B.1 with an empty supplement as the first inline element of this paragraph, to make sure that there is no funny whitespace introduced before figure references.

Here is a reference to a figure in a previous chapter: Figure 8.1. Notice the period after the reference, meaning the spacing around the reference works as intended.

Here is a reference to the same figure but with an alternative supplement: Kuvassa 8.1.

Table reference: Table 8.1.

Table reference with an alternative supplement: Taulukossa 8.1.

Table reference with an empty supplement: 8.1.

B.2 SECTION TESTS

A chapter reference: 1.

A chapter reference with an alternative supplement: Luvussa 1.

1 as the first element of this paragraph to test that there is no unnecessary whitespace before heading references.

B.3 EQUATION TESTS

Here is an equation reference: (B.1).

Reference equation in previous chapter: (8.1).

Check that supplements are ignored in the case of equation references: (B.1).

(B.1) as the first element of this paragraph. Again, no predecing whitespace should be generated.

B.4 CITATION TESTS

A citation test: [56].

A citation test with page numbers: [56, s. 1–2]

[56]: no preceding whitespace at the start of a paragraph.

B.5 THEOREM TESTS

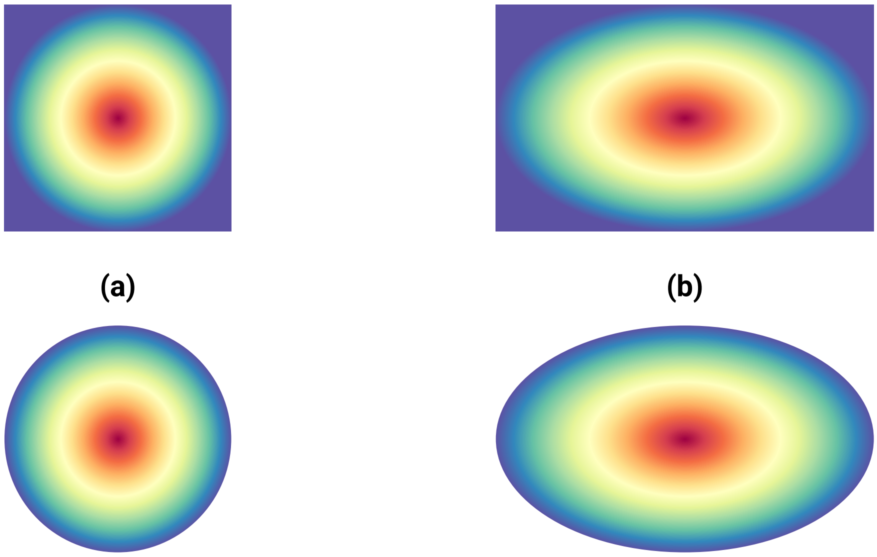
Referencing a theorem in this section: Theorem B.1.

Referencing a theorem box in another section: Example 8.1

Referencing a theorem box in another section in Finnish: Esimerkki 8.2

B.6 SUBFIGURE TESTS

Here we test referencing subfigures in a different section. These are the linear gradient in Figure 8.2.a, the radial gradient in Figure 8.2.b and the conic gradient in Figure 8.2.c.



(a) (b)

(c) (d)

Shapes B.1. A few different kinds of shapes. Shapes B.1.a contains a square, Shapes B.1.b a rectangle, Shapes B.1.c a circle and Shapes B.1.d an ellipse.

Shapes B.1 is here for testing subfigures inside of figures of arbitrary kinds.

Appendix C: Test appendix with mathematics in its heading: $\mathbf{f} = m\mathbf{a}$. Also some `code`.

This test appendix is here to test how the text sizes in the lists of figures and such behave.

Figure C.1. A test figure with mathematics in the caption: $\int_{\{x\}} f(x)\,\mathrm{d}x$.

The text sizes should be sensible, and mathematics should also scale to the surrounding text size.

C.1 A level 2 heading with mathematics: $\int_{\{x\}} f(x)\,\mathrm{d}x$

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

C.1.1 A level 3 heading with mathematics: $\int_{\{x\}} f(x)\,\mathrm{d}x$

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C.2 Another level 2 heading with mathematics: $\nabla \cdot \mathbf{j} = 0$ `more code`

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C.2.1 ANOTHER LEVEL 3 HEADING WITH MATHEMATICS: $\nabla \cdot \mathbf{j} = 0$

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C.3 A HEADING WITH `code`

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C.3.1 A HEADING WITH `more code`

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C.3.2 A HEADING WITH `even more code`

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